

# **SSN COLLEGE OF ENGINEERING**

## **Experiment 1: Working with Python packages-Numpy, Scipy, Scikit-Learn, Matplotlib**

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### **Aim**

To explore basic python libraries Numpy,Pandas,Scipy,ScikitLearn ,Matplotlib and perform EDA

### **Code**

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os
```

## FILE READING

```
In [2]: def readfile():
df=pd.read_csv("C:/Users/DHARSHINI/Downloads/sem5/machine-learning/expermien
df.columns=df.columns.str.strip()
df=df.reset_index(drop=True)
df=df.drop("Customer ID",axis=1)
df=df.drop("Name",axis=1)
print(df.head())
data_inspect(df)
```

## DATA STRUCTURE INSPECT

```
In [3]: def data_inspect_visual(df):
cols=df.select_dtypes(include=['int64','float64']).columns
for i in cols:
plt.figure(figsize=(5,5))
df[i].hist(bins=20)
plt.xlabel(i)
plt.ylabel("Frequency")
plt.show()
```

```
In [4]: def data_inspect(df):
print("SIZE:",df.size)
print("\n\nCOLUMNS\n",df.columns)
print("\n\nDATA TYPES\n",df.dtypes)
print("\n\nDESCRIBE\n",df.describe())
print("\n\nINFO\n",df.info())
data_inspect_visual(df)
missing_values(df)
return
```

## MISSING VALUES

```
In [5]: def missingvalues_heat_map(df):
plt.figure(figsize=(5,5))
sns.heatmap(df.isnull(),cbar=False,cmap="viridis")
plt.title("\nMISSING VALUES ")
plt.show()
```

```
In [6]: def missing_values(df):
print("\n\nNULL VALUES\n",df.isnull().sum())
print("\n\nNULL VALUES PERCENTAGE\n",(df.isnull().sum()/len(df))*100)

missingvalues_heat_map(df)
duplicate_values(df)
```

## DUPLICATE VALUES

```
In [7]: def duplicate_values(df):
        duplicate=df.duplicated().sum()
        print("\n\nDUPLICATES VALUES ",duplicate)
        if(duplicate>0):
            df=df.drop_duplicates()
            print("\nAfter Removing Duplicates : ",df.duplicated().sum())
        print("Calling categorical_features...")
        categorical_features(df)
        return df
```

## CATEGORICAL FEATURE PROFILING

```
In [8]: def value_count_plot(df,col):

        plt.figure(figsize=(8,5))
        sns.countplot(data=df,y=col)
        plt.title("VALUE COUNT PLOT ")
        plt.show()
```

```
In [9]: def categorical_features(df):
        cate=df.select_dtypes(include='object').columns

        for col in cate:
            print("UNIQUE VALUES IN THE ",col," : ",df[col].unique())
            value_count_plot(df,col)
        numerical_feature_analysis(df)
```

## NUMERICAL FEATURE PROFILING

```
In [10]: def numerical_feature_analysis(df):
        cols=df.select_dtypes(include=['int64','float64']).columns
        print("\n\nDESCRIBE\n",df.describe())

        print("\n\nSKEWNESS")
        for i in cols:
            print(f"{i}:{df[i].skew()}")

        linear_correlation(df)
```

## CORRELATION ANALYSIS

```
In [11]: def linear_correlation(df):
        plt.figure(figsize=(10,10))
        sns.heatmap(df.corr(numeric_only=True),annot=True,cmap='coolwarm')
        plt.title("LINEAR CORRELATION")
        plt.show()

        outliers(df)
```

# OUTLIERS VISUALS

```
In [12]: def outliers(df):  
          print("OUTLIERS VISUALS")  
          cate=df.select_dtypes(include="number").columns  
          for col in cate:  
              plt.figure(figsize=(5,5))  
              plt.boxplot(x=df[col])  
              plt.xlabel(col)  
              plt.show()
```

```
In [13]: readfile()
```

|   | Gender | Age | Income (USD) | Income Stability | Profession | \ |
|---|--------|-----|--------------|------------------|------------|---|
| 0 | F      | 56  | 1933.05      | Low              | Working    |   |
| 1 | M      | 32  | 4952.91      | Low              | Working    |   |
| 2 | F      | 65  | 988.19       | High             | Pensioner  |   |
| 3 | F      | 65  | NaN          | High             | Pensioner  |   |
| 4 | F      | 31  | 2614.77      | Low              | Working    |   |

|   | Type of Employment    | Location   | Loan Amount Request (USD) | \ |
|---|-----------------------|------------|---------------------------|---|
| 0 | Sales staff           | Semi-Urban | 72809.58                  |   |
| 1 | NaN                   | Semi-Urban | 46837.47                  |   |
| 2 | NaN                   | Semi-Urban | 45593.04                  |   |
| 3 | NaN                   | Rural      | 80057.92                  |   |
| 4 | High skill tech staff | Semi-Urban | 113858.89                 |   |

|   | Current Loan Expenses (USD) | Expense Type 1 | ... Credit Score | \ |
|---|-----------------------------|----------------|------------------|---|
| 0 | 241.08                      | N              | 809.44           |   |
| 1 | 495.81                      | N              | 780.40           |   |
| 2 | 171.95                      | N              | 833.15           |   |
| 3 | 298.54                      | N              | 832.70           |   |
| 4 | 491.41                      | N              | 745.55           |   |

|   | No. of Defaults | Has Active Credit Card | Property ID | Property Age | \ |
|---|-----------------|------------------------|-------------|--------------|---|
| 0 | 0               | NaN                    | 746         | 1933.05      |   |
| 1 | 0               | Unpossessed            | 608         | 4952.91      |   |
| 2 | 0               | Unpossessed            | 546         | 988.19       |   |
| 3 | 1               | Unpossessed            | 890         | NaN          |   |
| 4 | 1               | Active                 | 715         | 2614.77      |   |

|   | Property Type | Property Location | Co-Applicant | Property Price | \ |
|---|---------------|-------------------|--------------|----------------|---|
| 0 | 4             | Rural             | 1            | 119933.46      |   |
| 1 | 2             | Rural             | 1            | 54791.00       |   |
| 2 | 2             | Urban             | 0            | 72440.58       |   |
| 3 | 2             | Semi-Urban        | 1            | 121441.51      |   |
| 4 | 4             | Semi-Urban        | 1            | 208567.91      |   |

|   | Loan Sanction Amount (USD) |
|---|----------------------------|
| 0 | 54607.18                   |
| 1 | 37469.98                   |
| 2 | 36474.43                   |
| 3 | 56040.54                   |
| 4 | 74008.28                   |

[5 rows x 22 columns]

SIZE: 660000

#### COLUMNS

```
Index(['Gender', 'Age', 'Income (USD)', 'Income Stability', 'Profession',
      'Type of Employment', 'Location', 'Loan Amount Request (USD)',
      'Current Loan Expenses (USD)', 'Expense Type 1', 'Expense Type 2',
      'Dependents', 'Credit Score', 'No. of Defaults',
      'Has Active Credit Card', 'Property ID', 'Property Age',
      'Property Type', 'Property Location', 'Co-Applicant', 'Property Price',
      'Loan Sanction Amount (USD)'],
      dtype='object')
```

#### DATA TYPES

|        |        |
|--------|--------|
| Gender | object |
| Age    | int64  |

```

Income (USD)                float64
Income Stability             object
Profession                  object
Type of Employment          object
Location                    object
Loan Amount Request (USD)    float64
Current Loan Expenses (USD)  float64
Expense Type 1              object
Expense Type 2              object
Dependents                  float64
Credit Score                float64
No. of Defaults              int64
Has Active Credit Card       object
Property ID                  int64
Property Age                 float64
Property Type                int64
Property Location            object
Co-Applicant                 int64
Property Price               float64
Loan Sanction Amount (USD)   float64
dtype: object

```

## DESCRIBE

|       | Age          | Income (USD) | Loan Amount Request (USD) | \ |
|-------|--------------|--------------|---------------------------|---|
| count | 30000.000000 | 2.542400e+04 | 30000.000000              |   |
| mean  | 40.092300    | 2.630574e+03 | 88826.333855              |   |
| std   | 16.045129    | 1.126272e+04 | 59536.949605              |   |
| min   | 18.000000    | 3.777000e+02 | 6048.240000               |   |
| 25%   | 25.000000    | 1.650457e+03 | 41177.755000              |   |
| 50%   | 40.000000    | 2.222435e+03 | 75128.075000              |   |
| 75%   | 55.000000    | 3.090593e+03 | 119964.605000             |   |
| max   | 65.000000    | 1.777460e+06 | 621497.820000             |   |

|       | Current Loan Expenses (USD) | Dependents   | Credit Score | \ |
|-------|-----------------------------|--------------|--------------|---|
| count | 29828.000000                | 27507.000000 | 28297.000000 |   |
| mean  | 400.936876                  | 2.253027     | 739.885381   |   |
| std   | 242.545375                  | 0.951162     | 72.163846    |   |
| min   | -999.000000                 | 1.000000     | 580.000000   |   |
| 25%   | 247.667500                  | 2.000000     | 681.880000   |   |
| 50%   | 375.205000                  | 2.000000     | 739.820000   |   |
| 75%   | 521.292500                  | 3.000000     | 799.120000   |   |
| max   | 3840.880000                 | 14.000000    | 896.260000   |   |

|       | No. of Defaults | Property ID  | Property Age | Property Type | \ |
|-------|-----------------|--------------|--------------|---------------|---|
| count | 30000.000000    | 30000.000000 | 2.515000e+04 | 30000.000000  |   |
| mean  | 0.193933        | 501.934700   | 2.631119e+03 | 2.460067      |   |
| std   | 0.395384        | 288.158086   | 1.132268e+04 | 1.118562      |   |
| min   | 0.000000        | 1.000000     | 3.777000e+02 | 1.000000      |   |
| 25%   | 0.000000        | 251.000000   | 1.650450e+03 | 1.000000      |   |
| 50%   | 0.000000        | 504.000000   | 2.223250e+03 | 2.000000      |   |
| 75%   | 0.000000        | 751.000000   | 3.091408e+03 | 3.000000      |   |
| max   | 1.000000        | 999.000000   | 1.777460e+06 | 4.000000      |   |

|       | Co-Applicant | Property Price | Loan Sanction Amount (USD) |
|-------|--------------|----------------|----------------------------|
| count | 30000.000000 | 3.000000e+04   | 29660.000000               |
| mean  | -4.743867    | 1.317597e+05   | 47649.342208               |
| std   | 74.614593    | 9.354955e+04   | 48221.146686               |
| min   | -999.000000  | -9.990000e+02  | -999.000000                |
| 25%   | 1.000000     | 6.057216e+04   | 0.000000                   |

```

50%      1.000000      1.099936e+05      35209.395000
75%      1.000000      1.788807e+05      74261.250000
max       1.000000      1.077967e+06      481907.320000

```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 30000 entries, 0 to 29999
```

```
Data columns (total 22 columns):
```

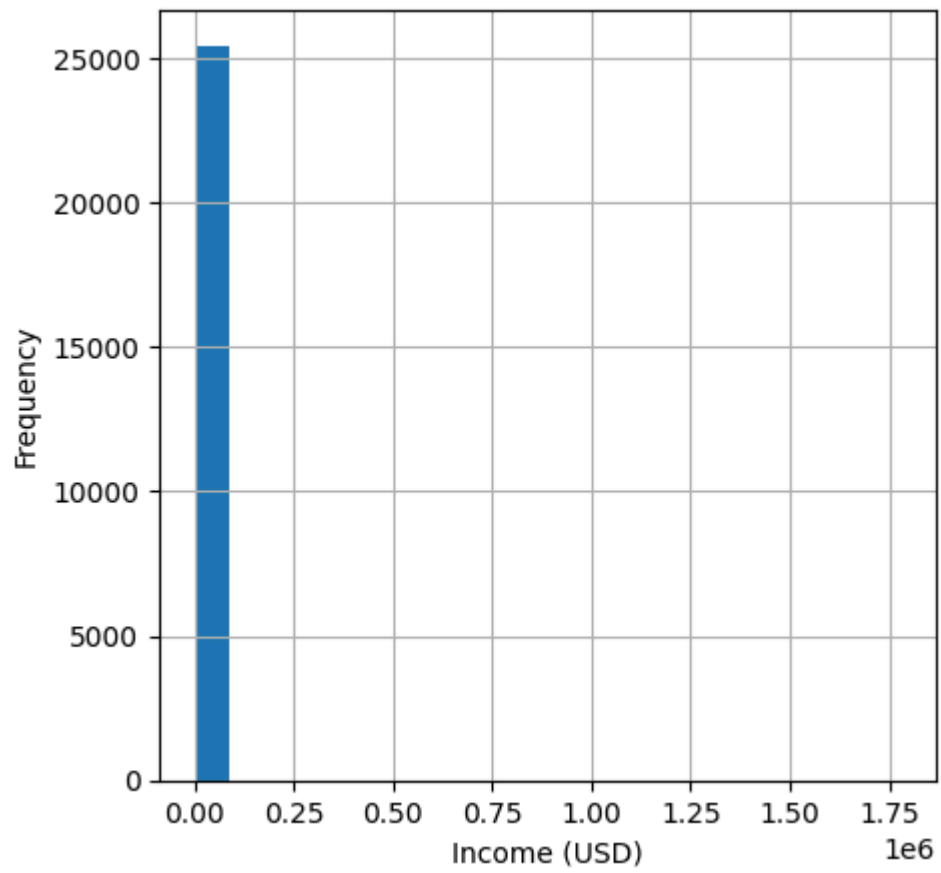
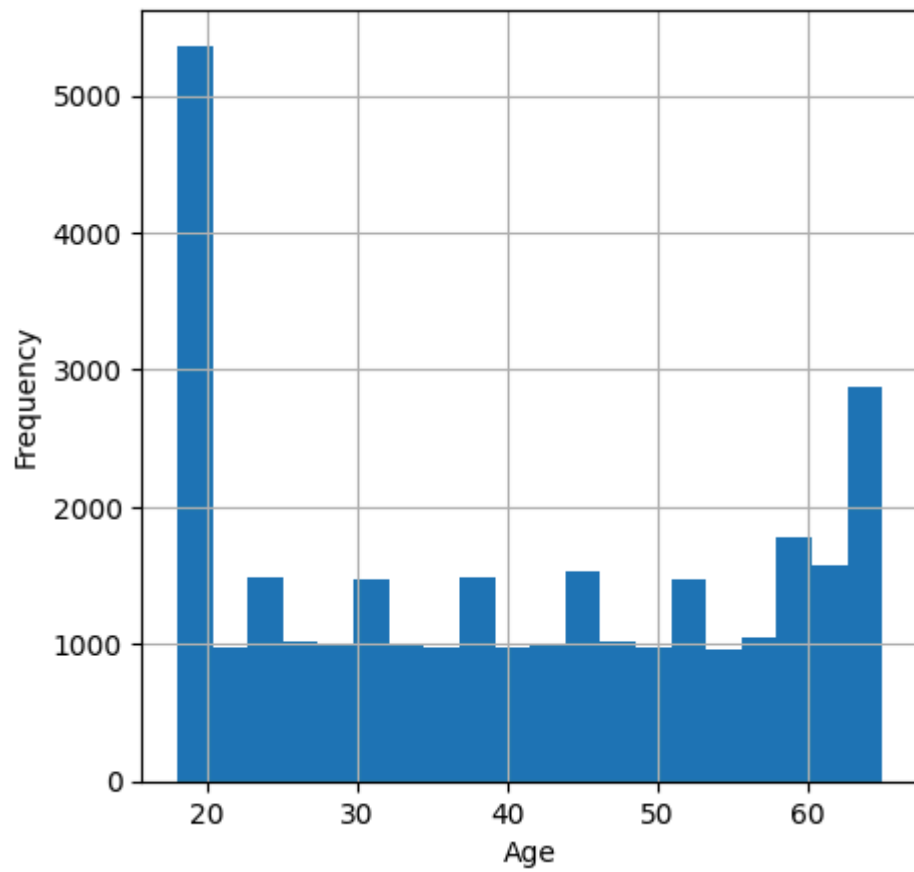
| #  | Column                      | Non-Null Count | Dtype   |
|----|-----------------------------|----------------|---------|
| 0  | Gender                      | 29947 non-null | object  |
| 1  | Age                         | 30000 non-null | int64   |
| 2  | Income (USD)                | 25424 non-null | float64 |
| 3  | Income Stability            | 28317 non-null | object  |
| 4  | Profession                  | 30000 non-null | object  |
| 5  | Type of Employment          | 22730 non-null | object  |
| 6  | Location                    | 30000 non-null | object  |
| 7  | Loan Amount Request (USD)   | 30000 non-null | float64 |
| 8  | Current Loan Expenses (USD) | 29828 non-null | float64 |
| 9  | Expense Type 1              | 30000 non-null | object  |
| 10 | Expense Type 2              | 30000 non-null | object  |
| 11 | Dependents                  | 27507 non-null | float64 |
| 12 | Credit Score                | 28297 non-null | float64 |
| 13 | No. of Defaults             | 30000 non-null | int64   |
| 14 | Has Active Credit Card      | 28434 non-null | object  |
| 15 | Property ID                 | 30000 non-null | int64   |
| 16 | Property Age                | 25150 non-null | float64 |
| 17 | Property Type               | 30000 non-null | int64   |
| 18 | Property Location           | 29644 non-null | object  |
| 19 | Co-Applicant                | 30000 non-null | int64   |
| 20 | Property Price              | 30000 non-null | float64 |
| 21 | Loan Sanction Amount (USD)  | 29660 non-null | float64 |

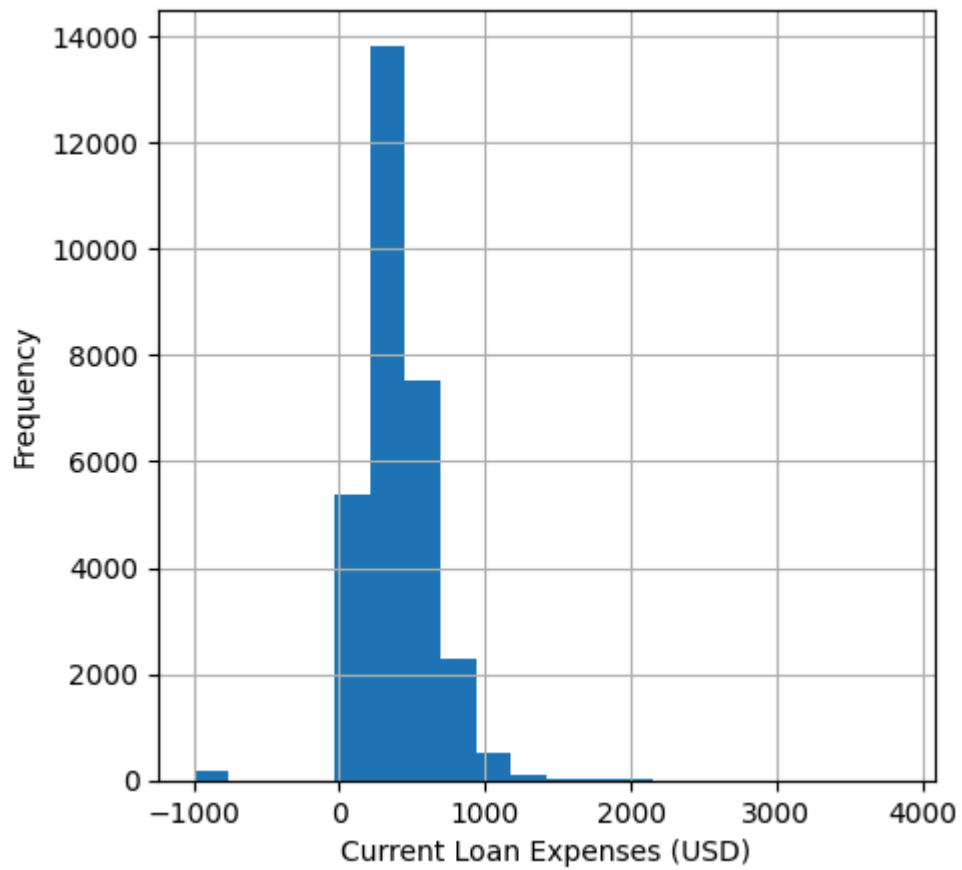
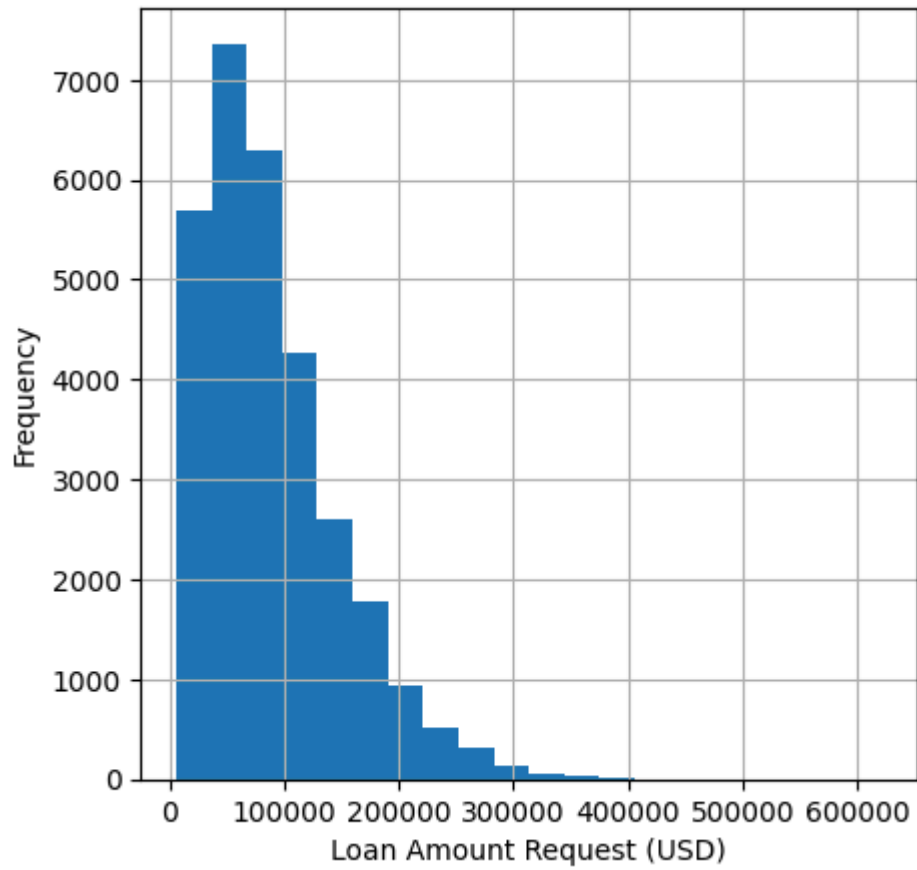
```
dtypes: float64(8), int64(5), object(9)
```

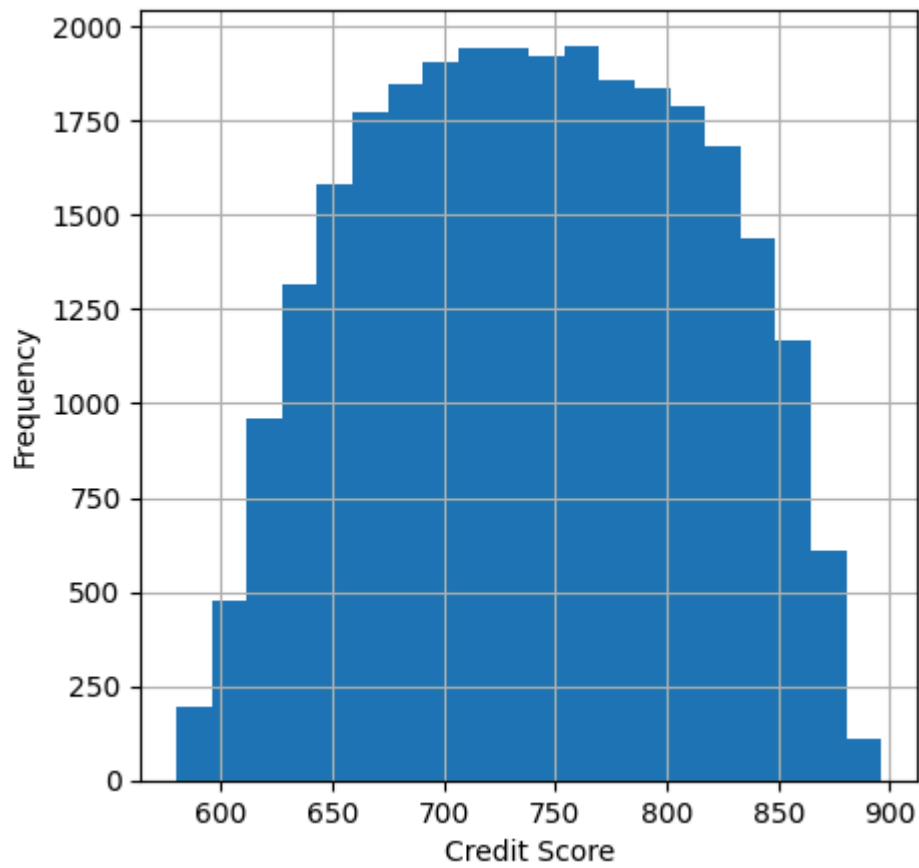
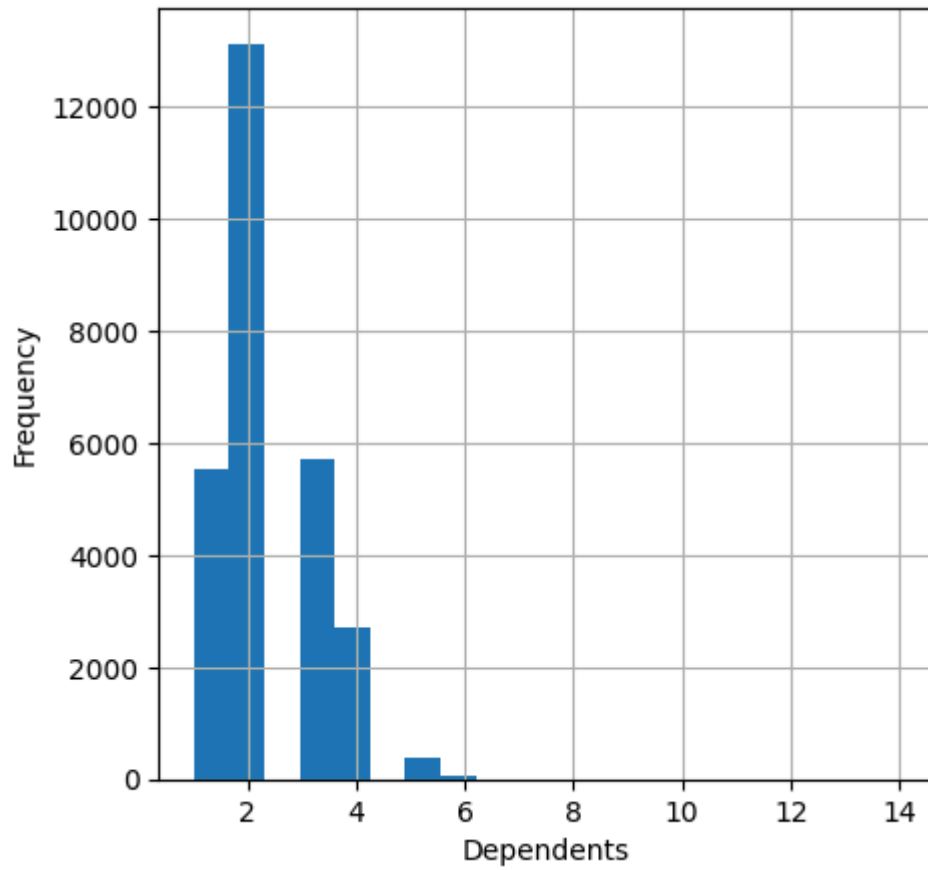
```
memory usage: 5.0+ MB
```

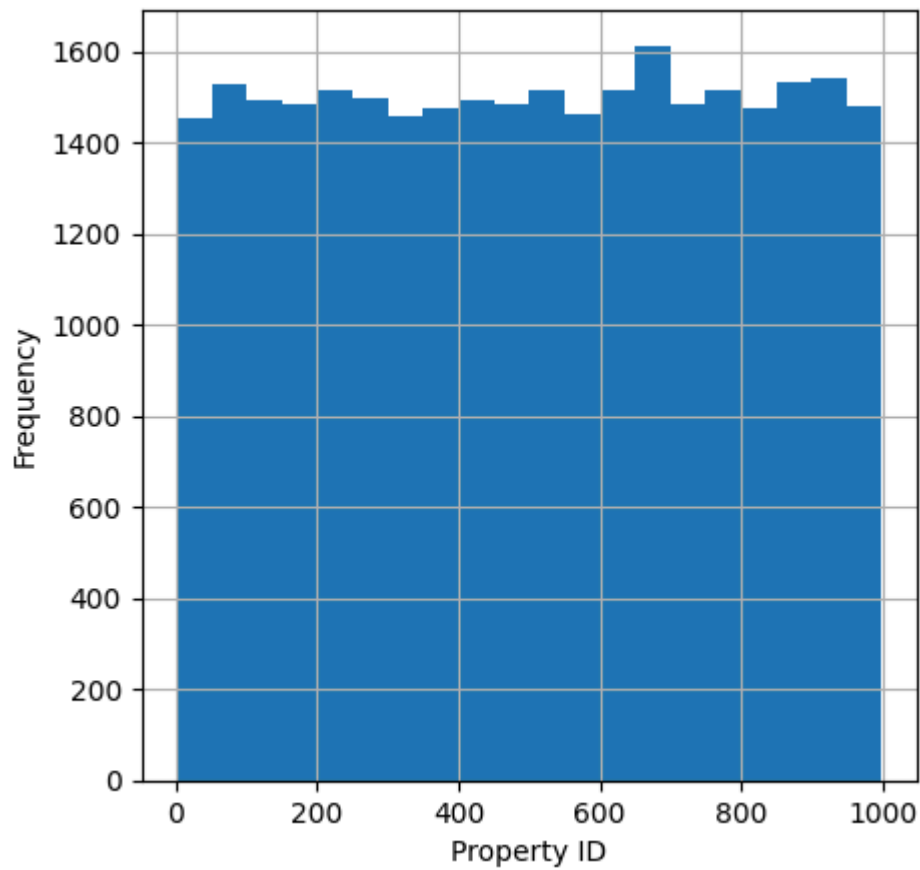
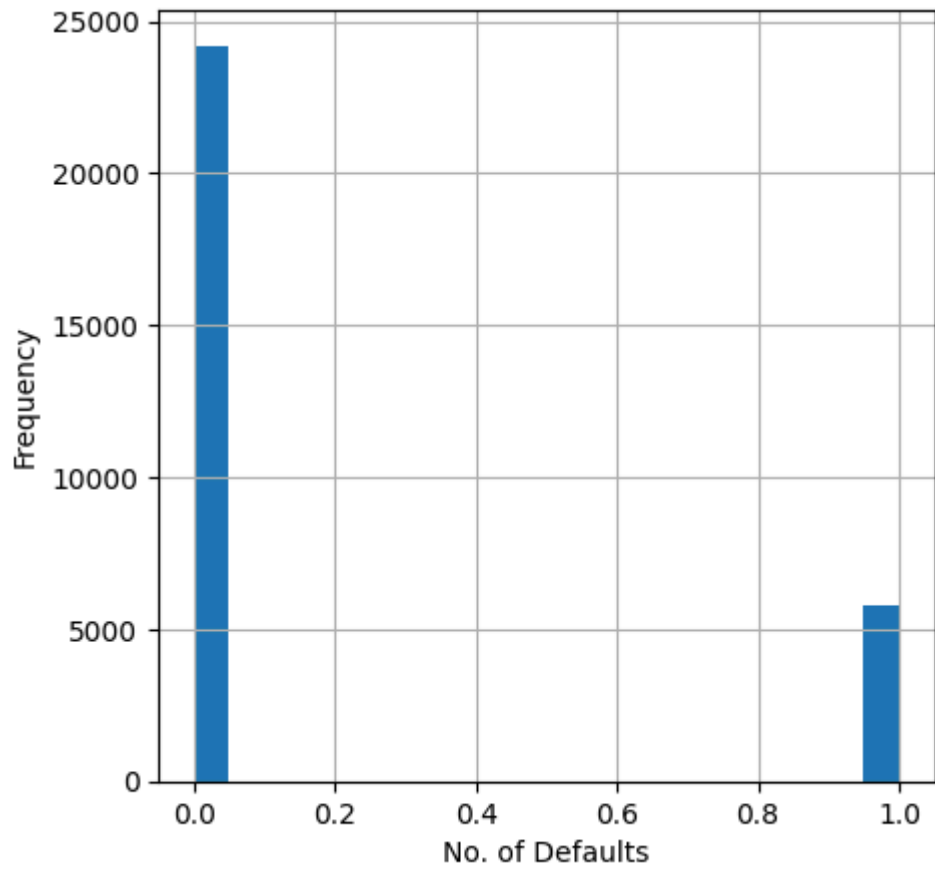
```
INFO
```

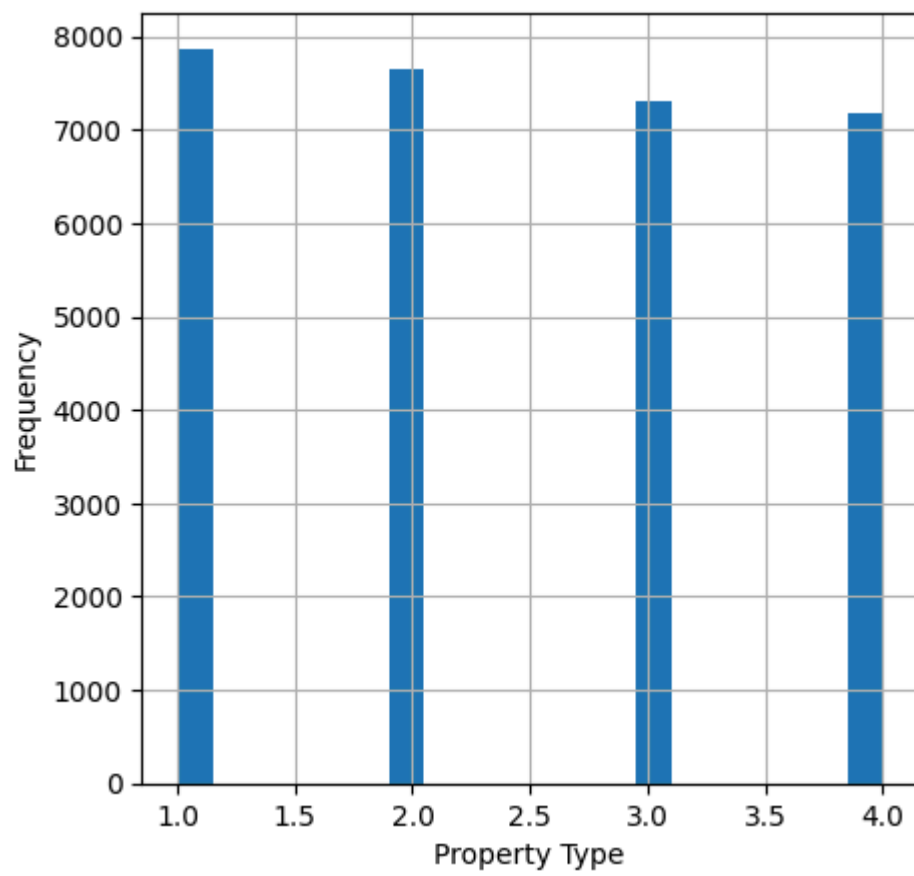
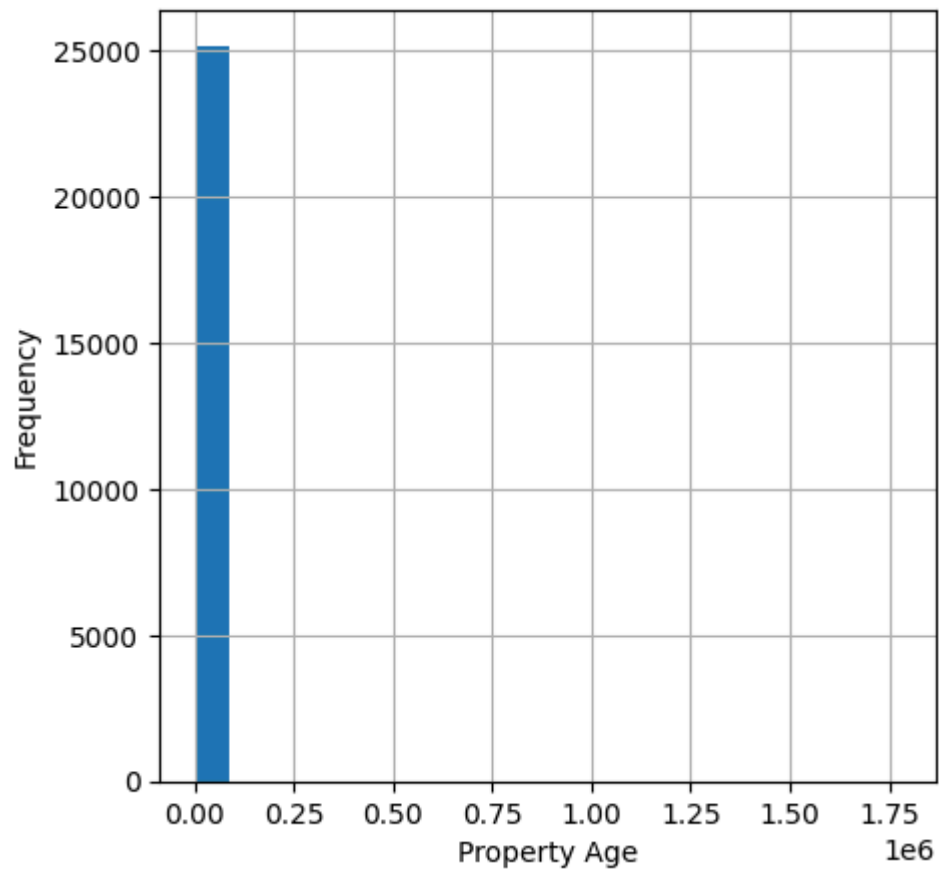
```
None
```

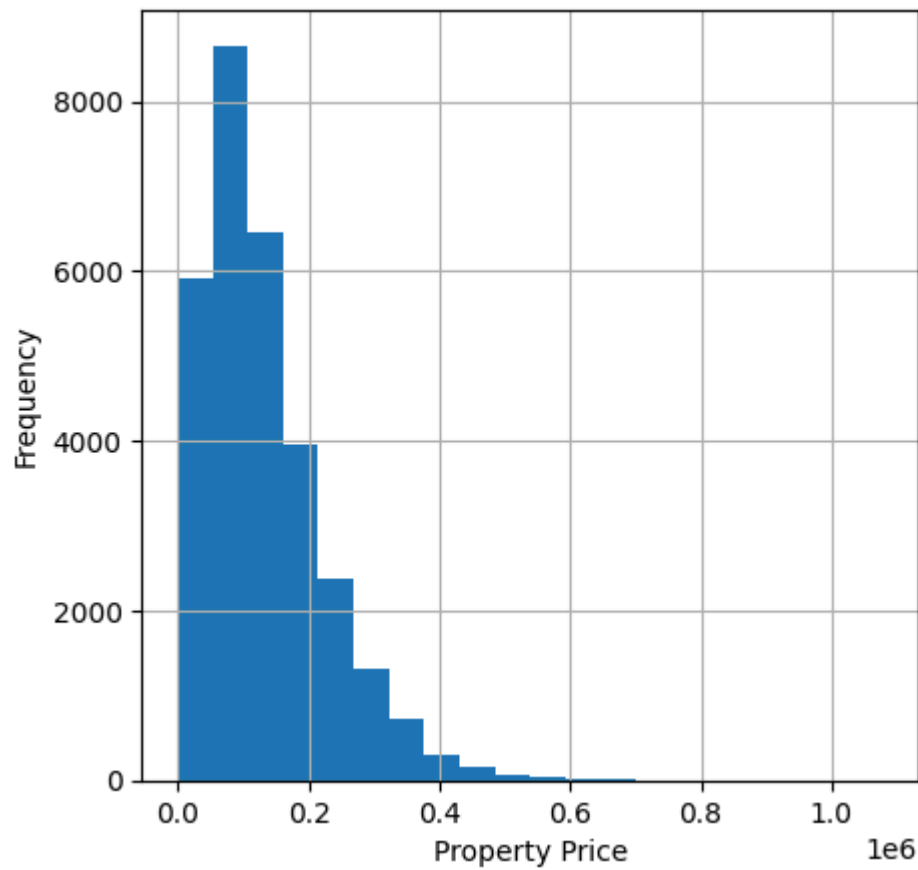
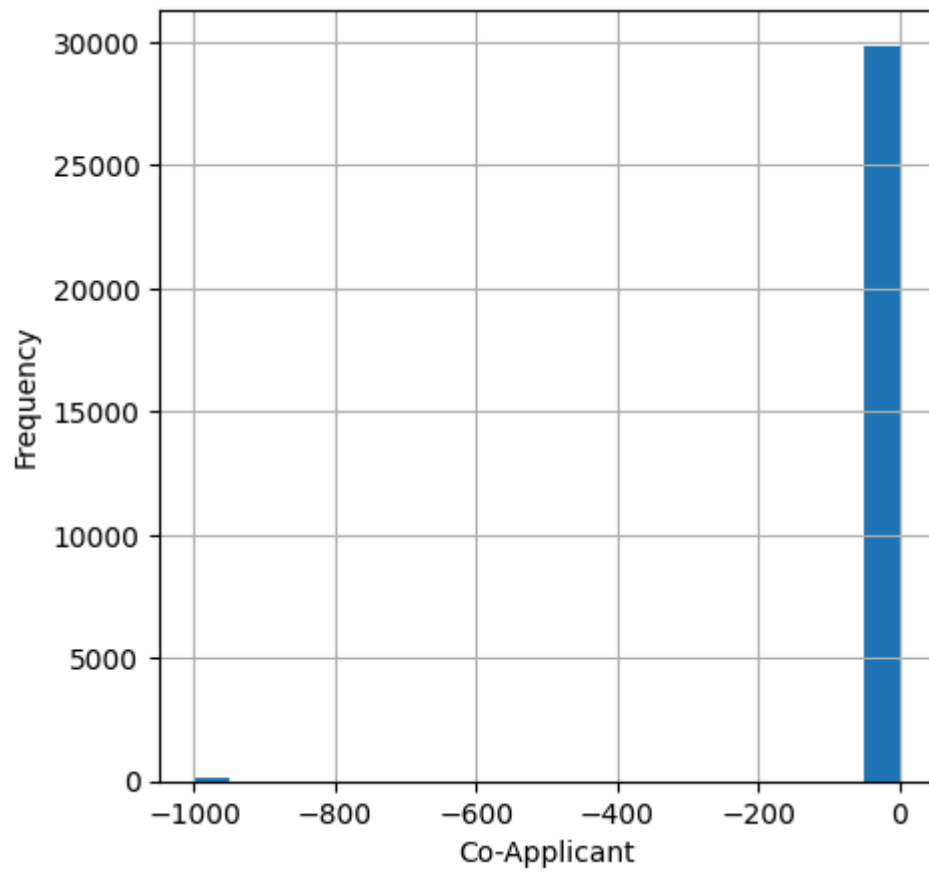


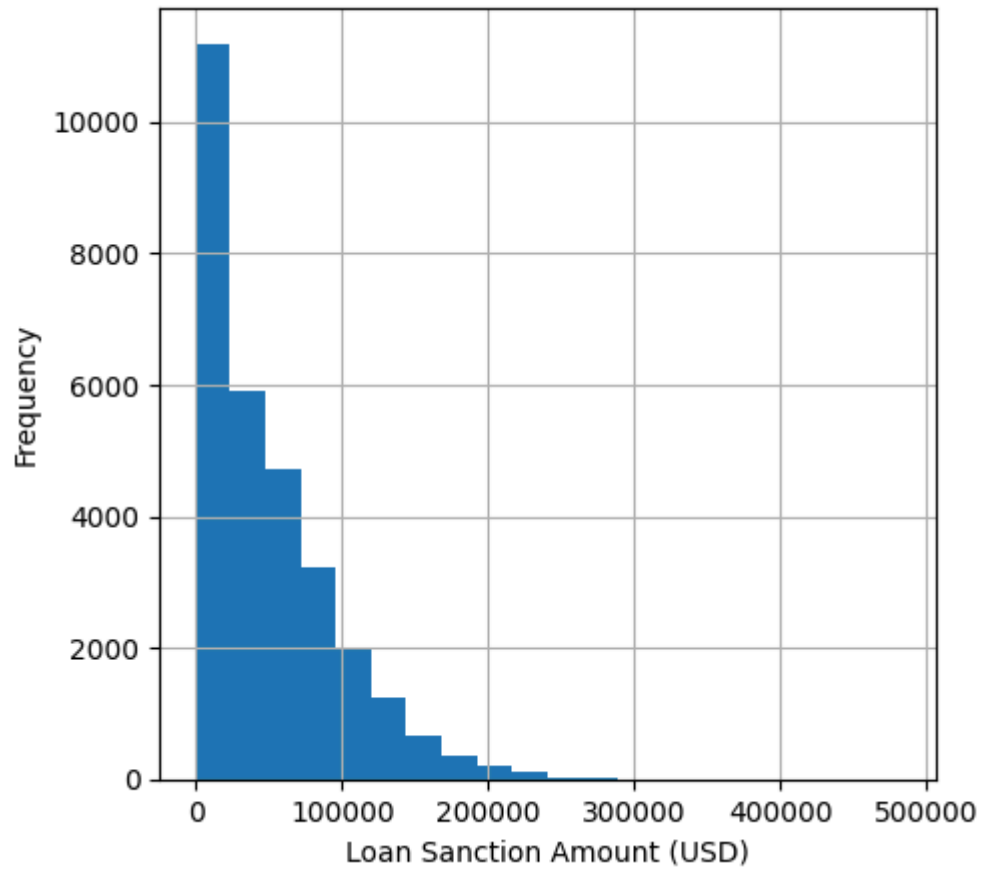










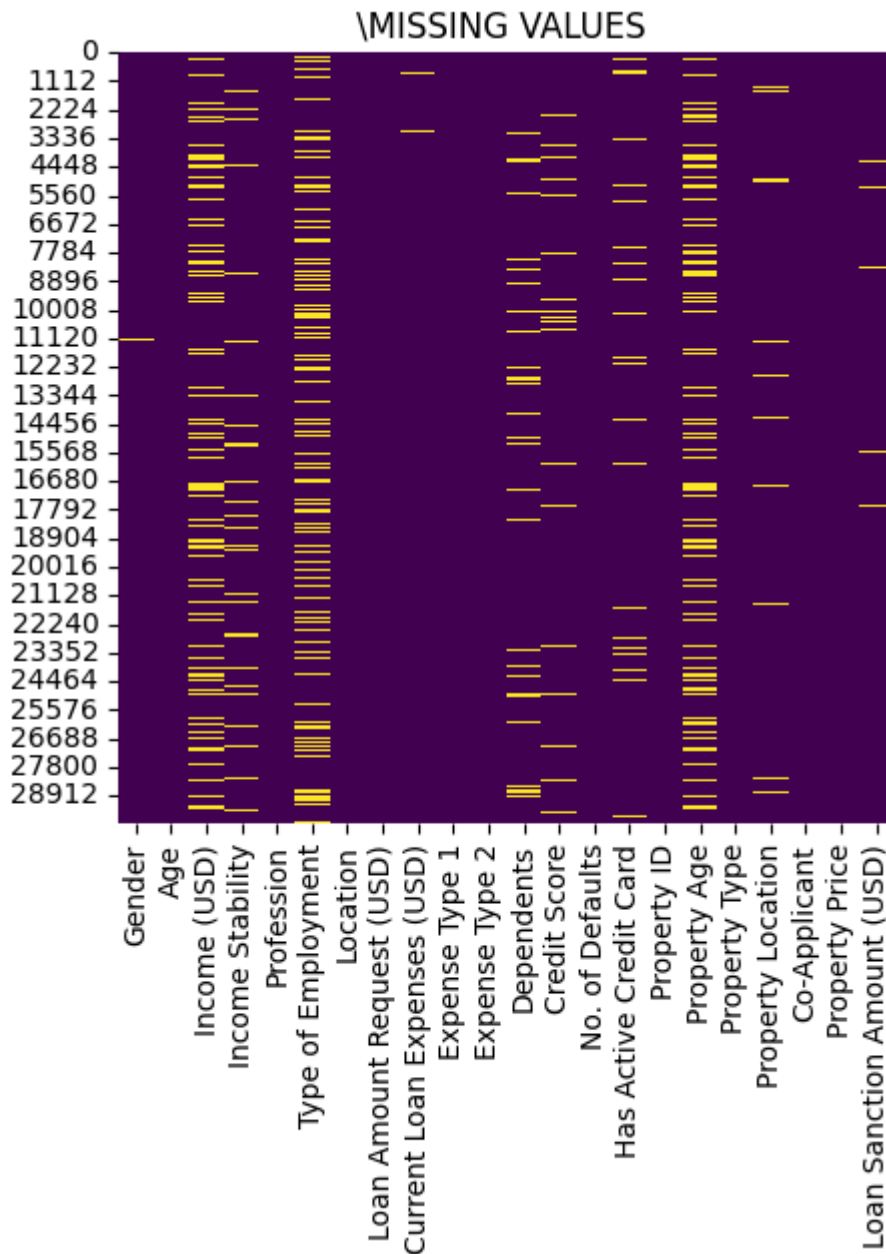


## NULL VALUES

|                             |       |
|-----------------------------|-------|
| Gender                      | 53    |
| Age                         | 0     |
| Income (USD)                | 4576  |
| Income Stability            | 1683  |
| Profession                  | 0     |
| Type of Employment          | 7270  |
| Location                    | 0     |
| Loan Amount Request (USD)   | 0     |
| Current Loan Expenses (USD) | 172   |
| Expense Type 1              | 0     |
| Expense Type 2              | 0     |
| Dependents                  | 2493  |
| Credit Score                | 1703  |
| No. of Defaults             | 0     |
| Has Active Credit Card      | 1566  |
| Property ID                 | 0     |
| Property Age                | 4850  |
| Property Type               | 0     |
| Property Location           | 356   |
| Co-Applicant                | 0     |
| Property Price              | 0     |
| Loan Sanction Amount (USD)  | 340   |
| dtype:                      | int64 |

## NULL VALUES PERCENTAGE

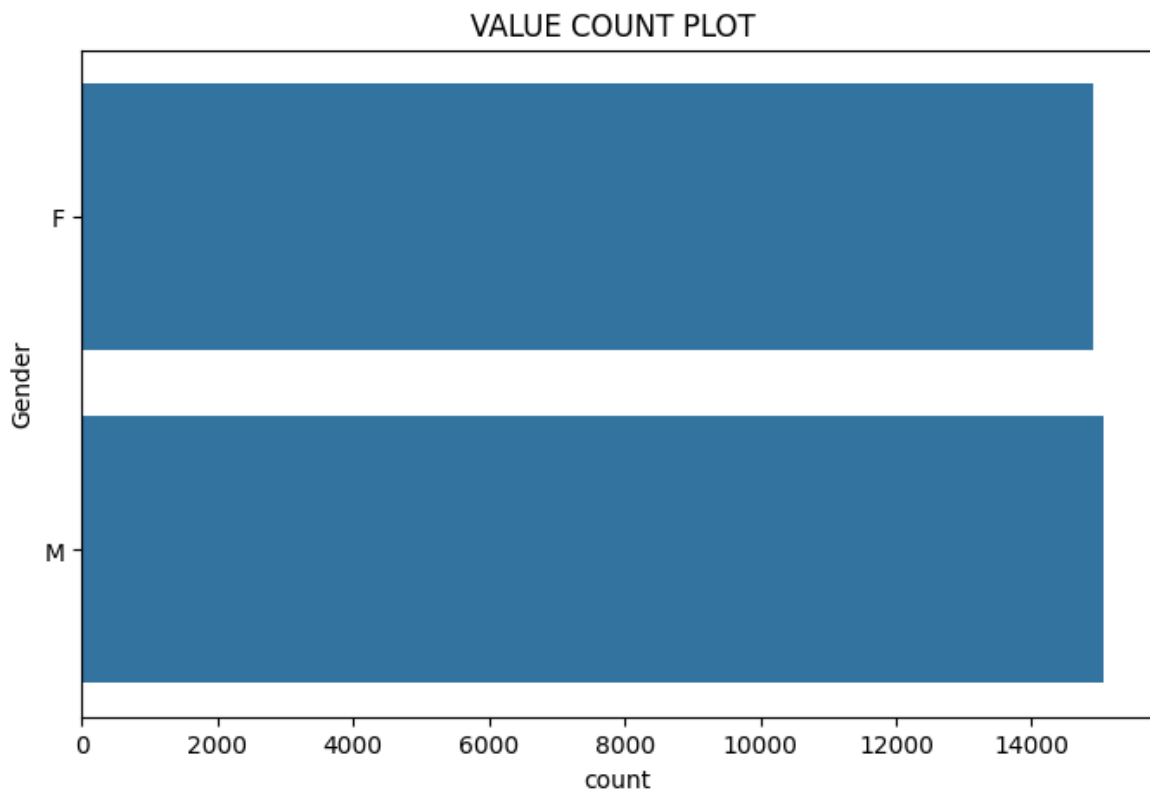
|                             |           |
|-----------------------------|-----------|
| Gender                      | 0.176667  |
| Age                         | 0.000000  |
| Income (USD)                | 15.253333 |
| Income Stability            | 5.610000  |
| Profession                  | 0.000000  |
| Type of Employment          | 24.233333 |
| Location                    | 0.000000  |
| Loan Amount Request (USD)   | 0.000000  |
| Current Loan Expenses (USD) | 0.573333  |
| Expense Type 1              | 0.000000  |
| Expense Type 2              | 0.000000  |
| Dependents                  | 8.310000  |
| Credit Score                | 5.676667  |
| No. of Defaults             | 0.000000  |
| Has Active Credit Card      | 5.220000  |
| Property ID                 | 0.000000  |
| Property Age                | 16.166667 |
| Property Type               | 0.000000  |
| Property Location           | 1.186667  |
| Co-Applicant                | 0.000000  |
| Property Price              | 0.000000  |
| Loan Sanction Amount (USD)  | 1.133333  |
| dtype:                      | float64   |



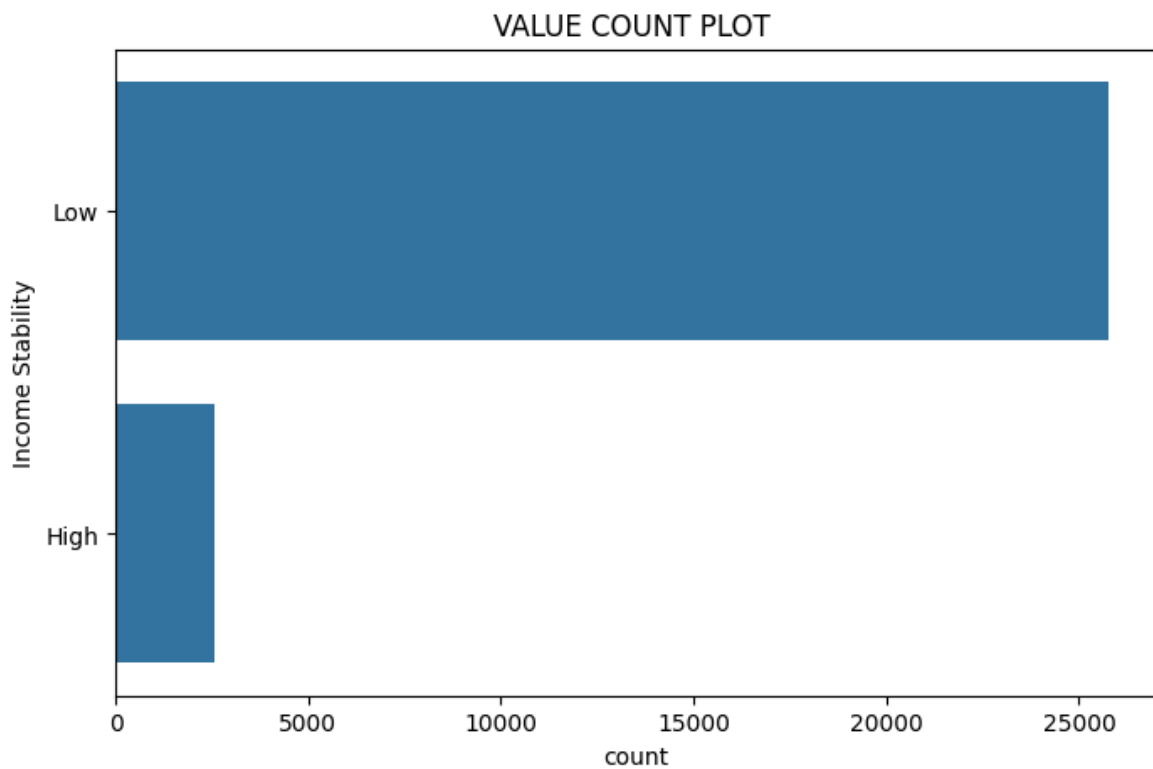
DUPLICATES VALUES 0

Calling categorical\_features...

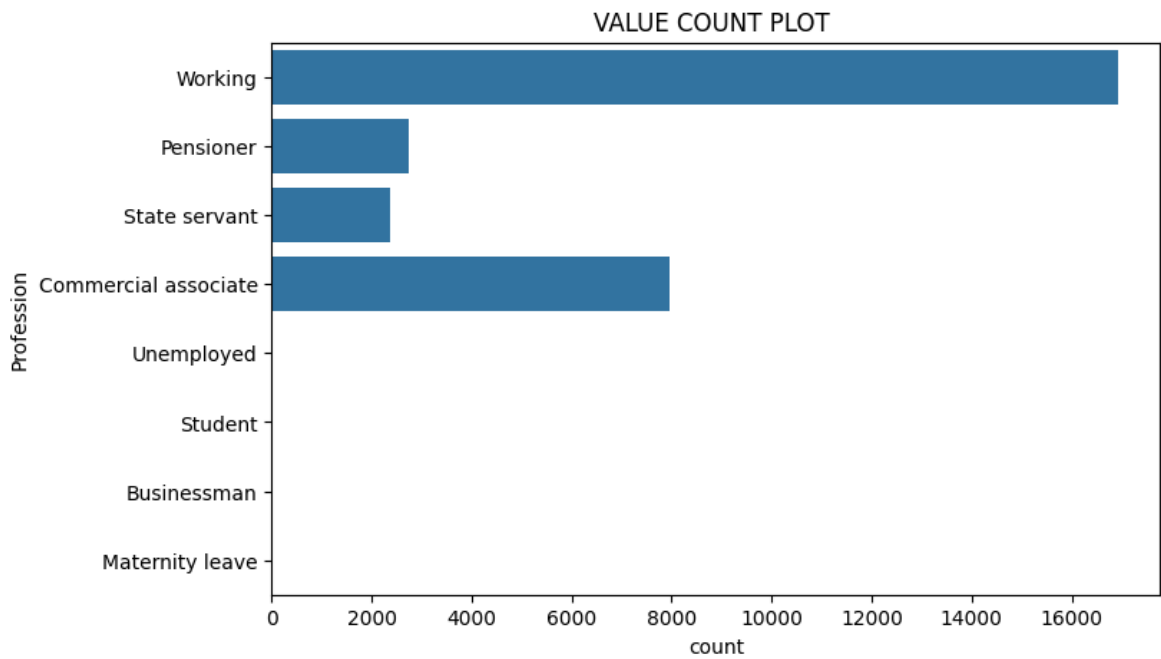
UNIQUE VALUES IN THE Gender : ['F' 'M' nan]



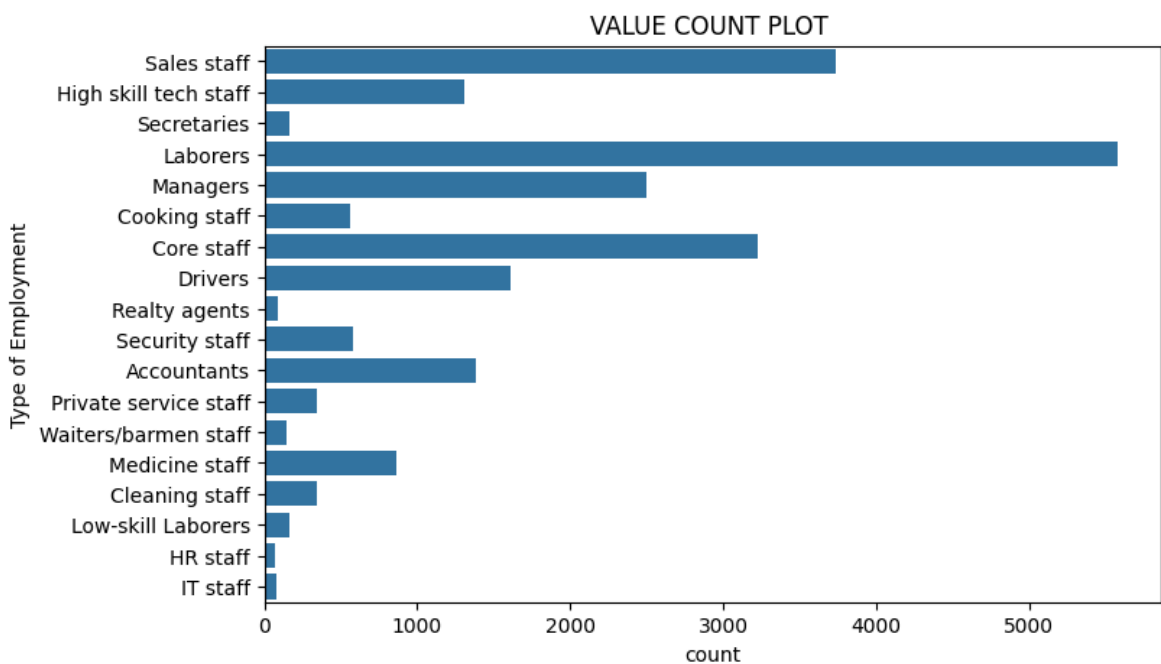
UNIQUE VALUES IN THE Income Stability : ['Low' 'High' nan]



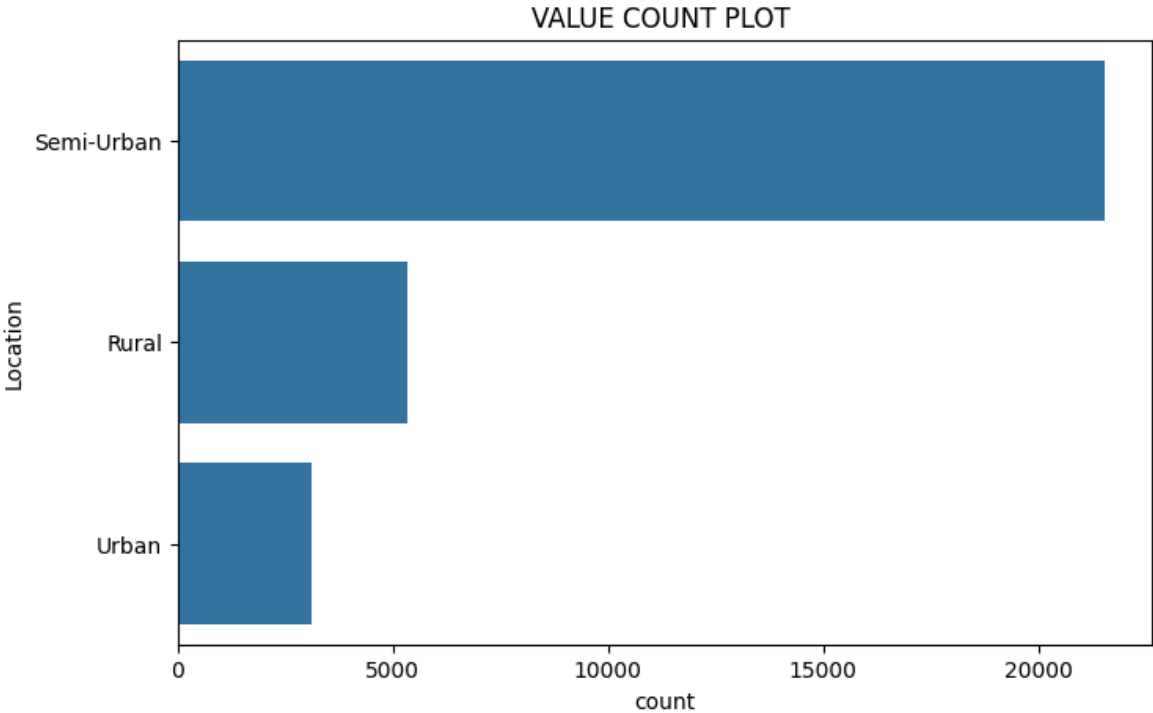
UNIQUE VALUES IN THE Profession : ['Working' 'Pensioner' 'State servant' 'Commercial associate' 'Unemployed' 'Student' 'Businessman' 'Maternity leave']



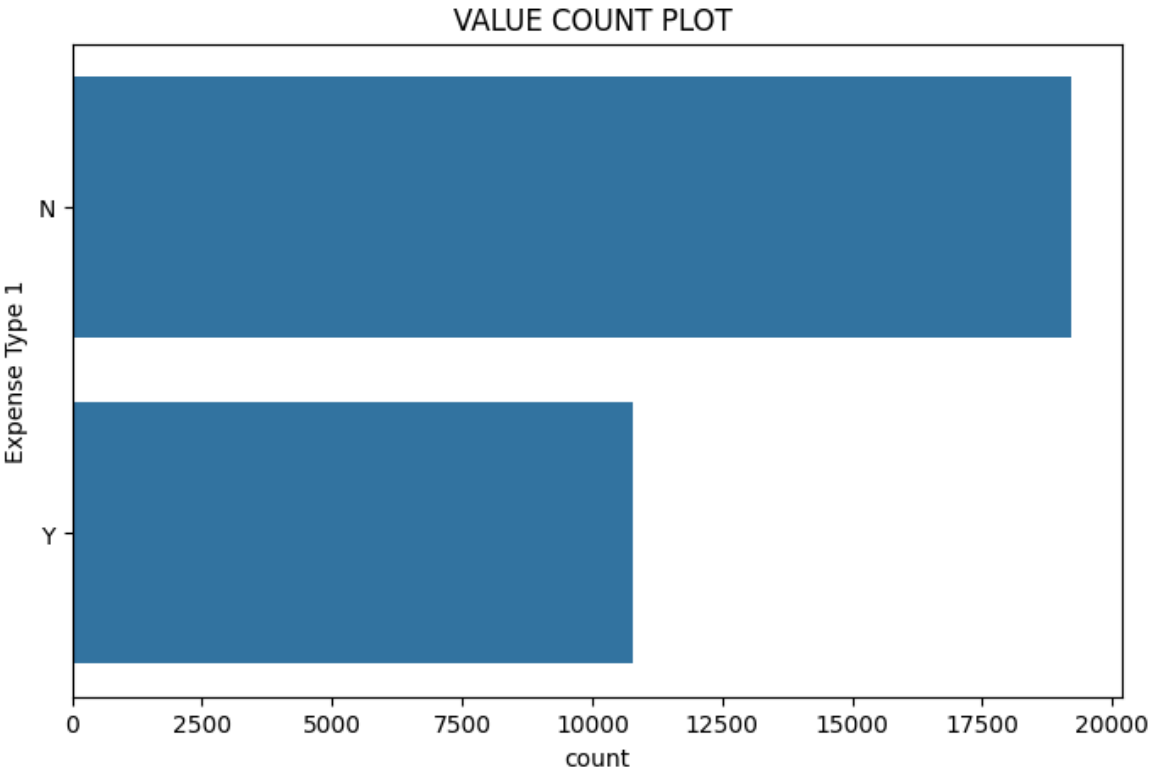
UNIQUE VALUES IN THE Type of Employment : ['Sales staff' nan 'High skill tech staff' 'Secretaries' 'Laborers' 'Managers' 'Cooking staff' 'Core staff' 'Drivers' 'Realty agents' 'Security staff' 'Accountants' 'Private service staff' 'Waiters/barmen staff' 'Medicine staff' 'Cleaning staff' 'Low-skill Laborers' 'HR staff' 'IT staff']



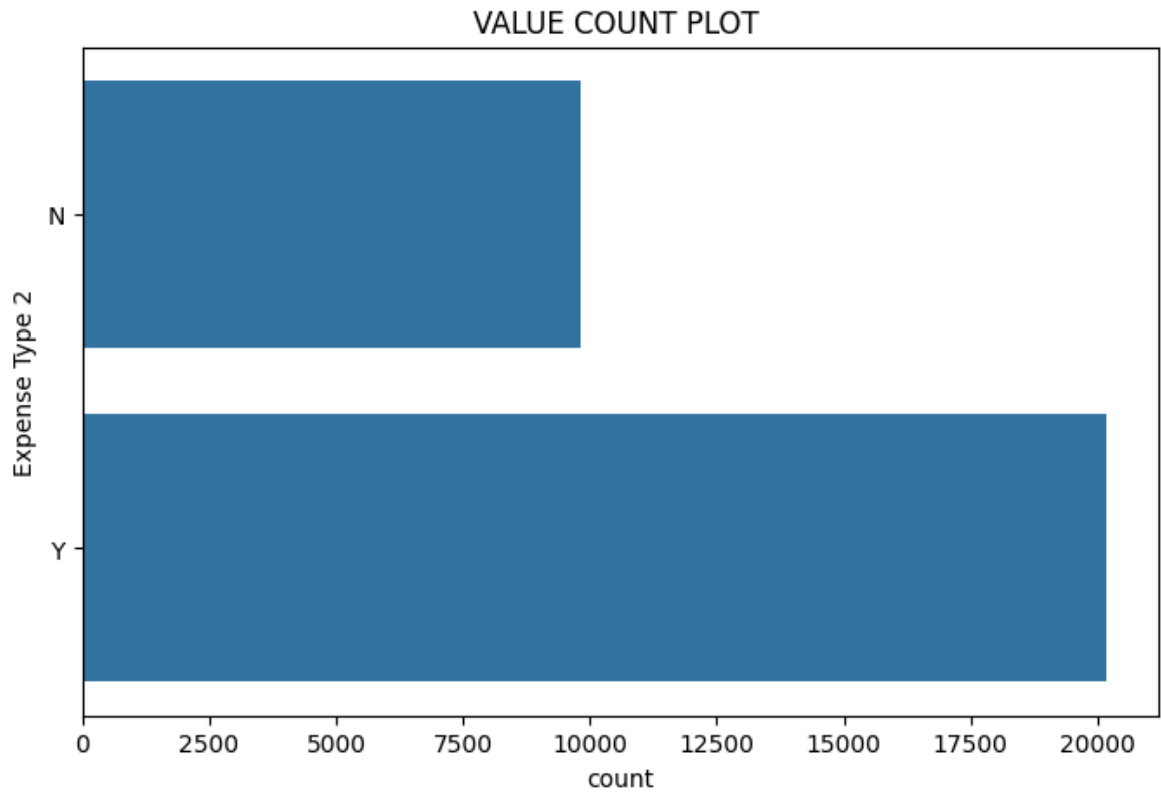
UNIQUE VALUES IN THE Location : ['Semi-Urban' 'Rural' 'Urban']



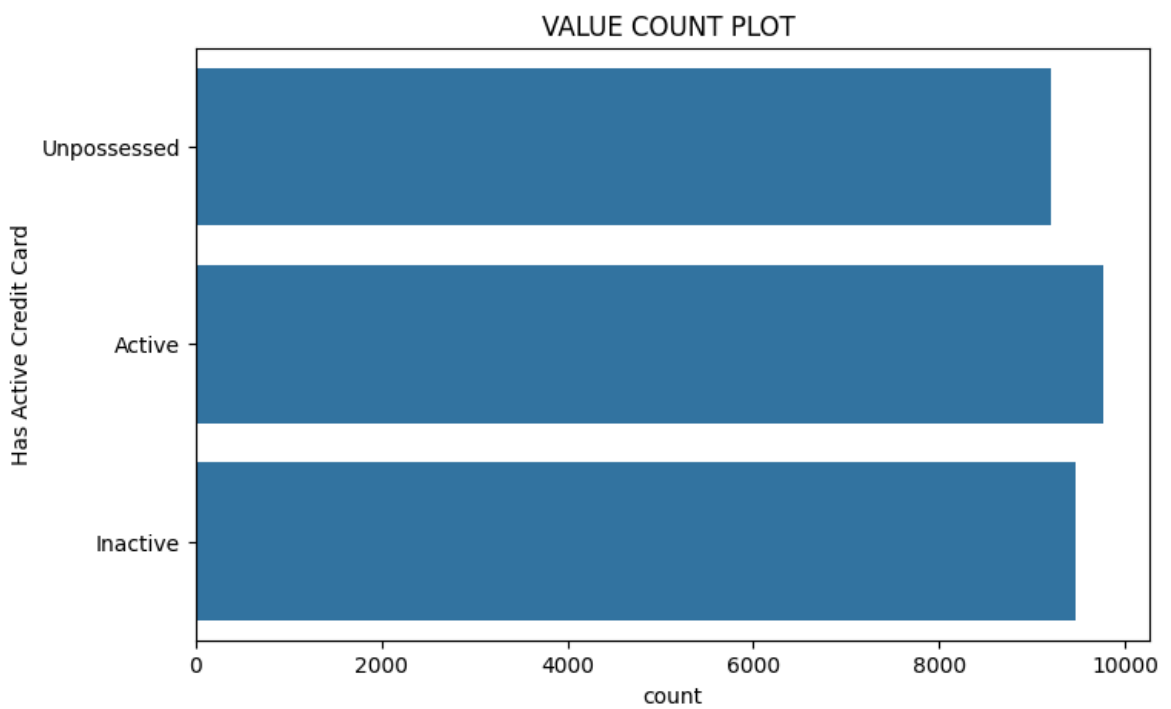
UNIQUE VALUES IN THE Expense Type 1 : ['N' 'Y']



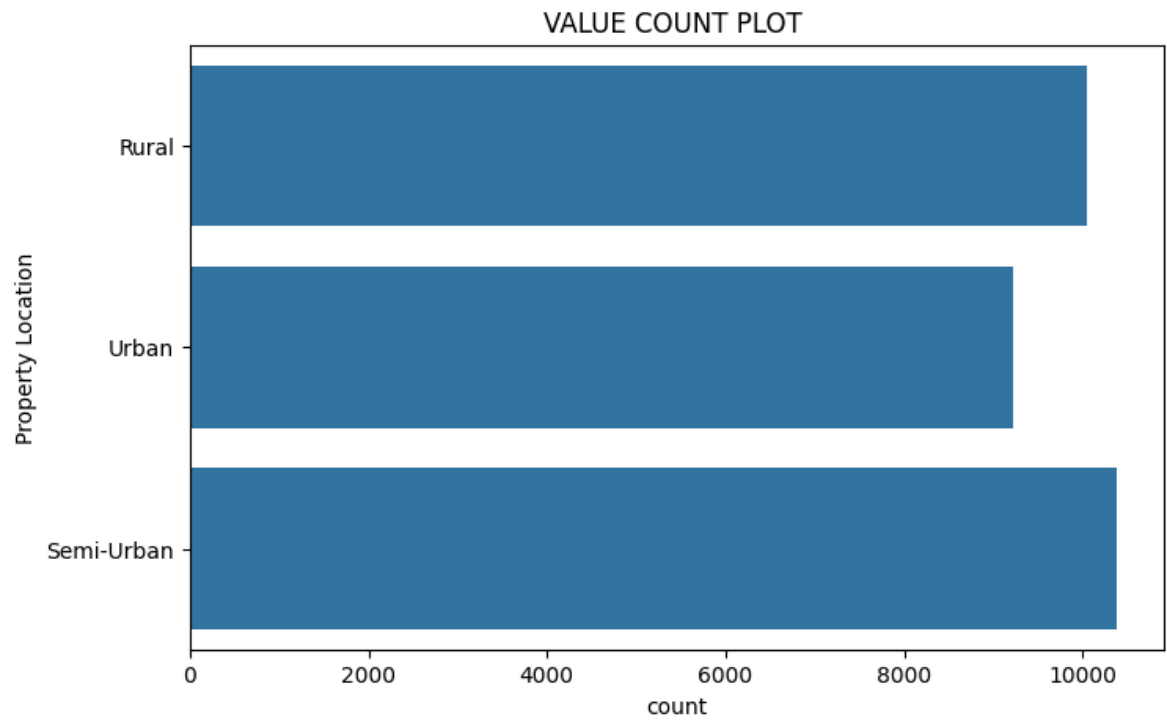
UNIQUE VALUES IN THE Expense Type 2 : ['N' 'Y']



UNIQUE VALUES IN THE Has Active Credit Card : [nan 'Unpossessed' 'Active' 'Inactive']



UNIQUE VALUES IN THE Property Location : ['Rural' 'Urban' 'Semi-Urban' nan]



## DESCRIBE

|       | Age          | Income (USD) | Loan Amount Request (USD) | \ |
|-------|--------------|--------------|---------------------------|---|
| count | 30000.000000 | 2.542400e+04 | 30000.000000              |   |
| mean  | 40.092300    | 2.630574e+03 | 88826.333855              |   |
| std   | 16.045129    | 1.126272e+04 | 59536.949605              |   |
| min   | 18.000000    | 3.777000e+02 | 6048.240000               |   |
| 25%   | 25.000000    | 1.650457e+03 | 41177.755000              |   |
| 50%   | 40.000000    | 2.22435e+03  | 75128.075000              |   |
| 75%   | 55.000000    | 3.090593e+03 | 119964.605000             |   |
| max   | 65.000000    | 1.777460e+06 | 621497.820000             |   |

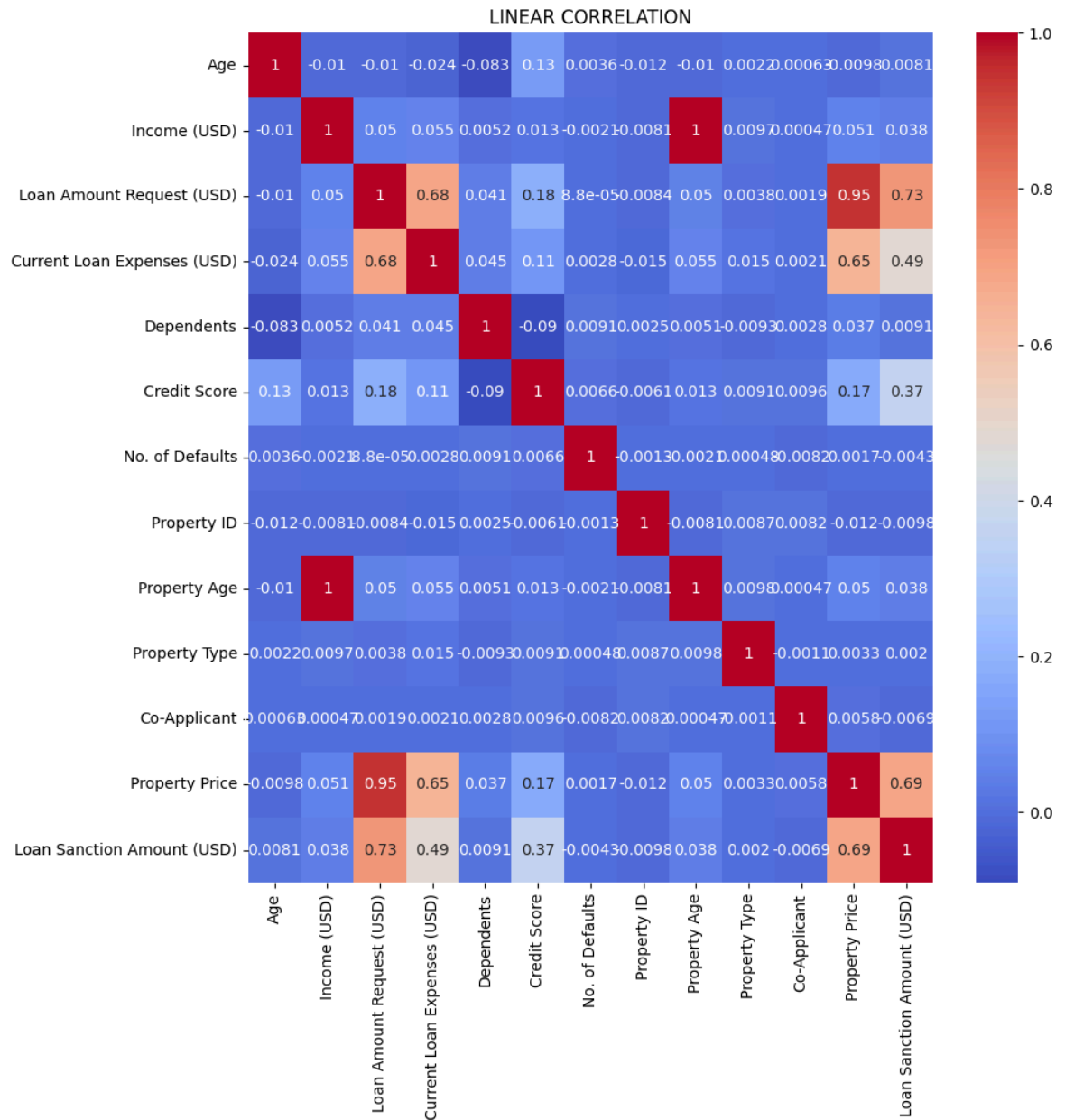
|       | Current Loan Expenses (USD) | Dependents   | Credit Score | \ |
|-------|-----------------------------|--------------|--------------|---|
| count | 29828.000000                | 27507.000000 | 28297.000000 |   |
| mean  | 400.936876                  | 2.253027     | 739.885381   |   |
| std   | 242.545375                  | 0.951162     | 72.163846    |   |
| min   | -999.000000                 | 1.000000     | 580.000000   |   |
| 25%   | 247.667500                  | 2.000000     | 681.880000   |   |
| 50%   | 375.205000                  | 2.000000     | 739.820000   |   |
| 75%   | 521.292500                  | 3.000000     | 799.120000   |   |
| max   | 3840.880000                 | 14.000000    | 896.260000   |   |

|       | No. of Defaults | Property ID  | Property Age | Property Type | \ |
|-------|-----------------|--------------|--------------|---------------|---|
| count | 30000.000000    | 30000.000000 | 2.515000e+04 | 30000.000000  |   |
| mean  | 0.193933        | 501.934700   | 2.631119e+03 | 2.460067      |   |
| std   | 0.395384        | 288.158086   | 1.132268e+04 | 1.118562      |   |
| min   | 0.000000        | 1.000000     | 3.777000e+02 | 1.000000      |   |
| 25%   | 0.000000        | 251.000000   | 1.650450e+03 | 1.000000      |   |
| 50%   | 0.000000        | 504.000000   | 2.223250e+03 | 2.000000      |   |
| 75%   | 0.000000        | 751.000000   | 3.091408e+03 | 3.000000      |   |
| max   | 1.000000        | 999.000000   | 1.777460e+06 | 4.000000      |   |

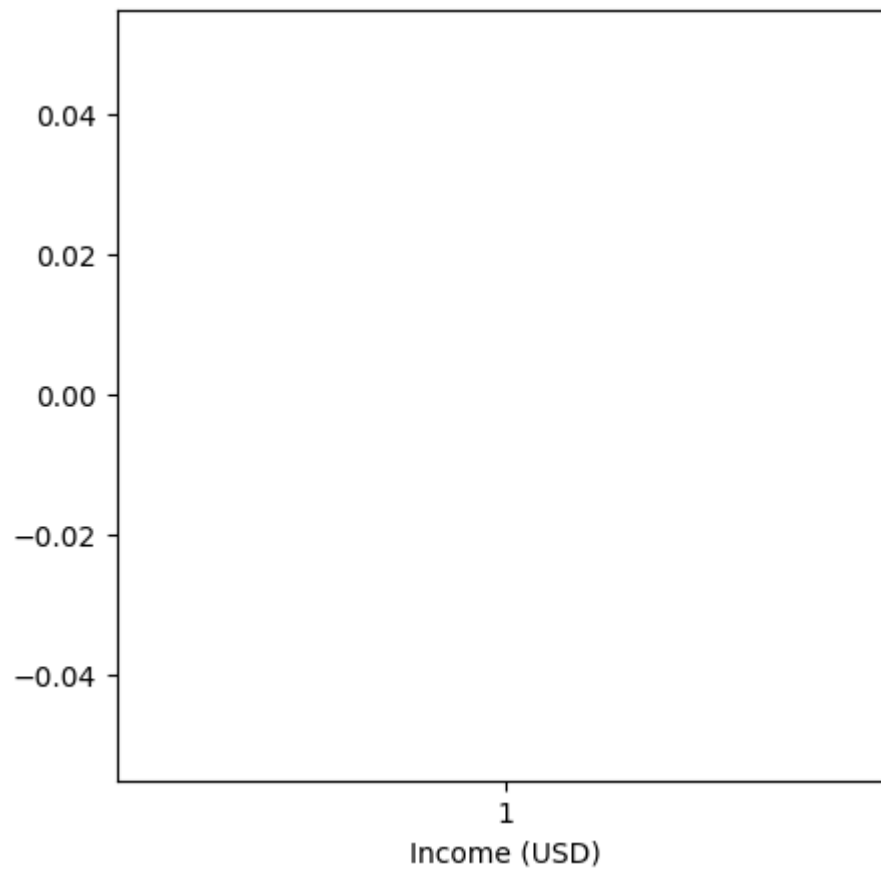
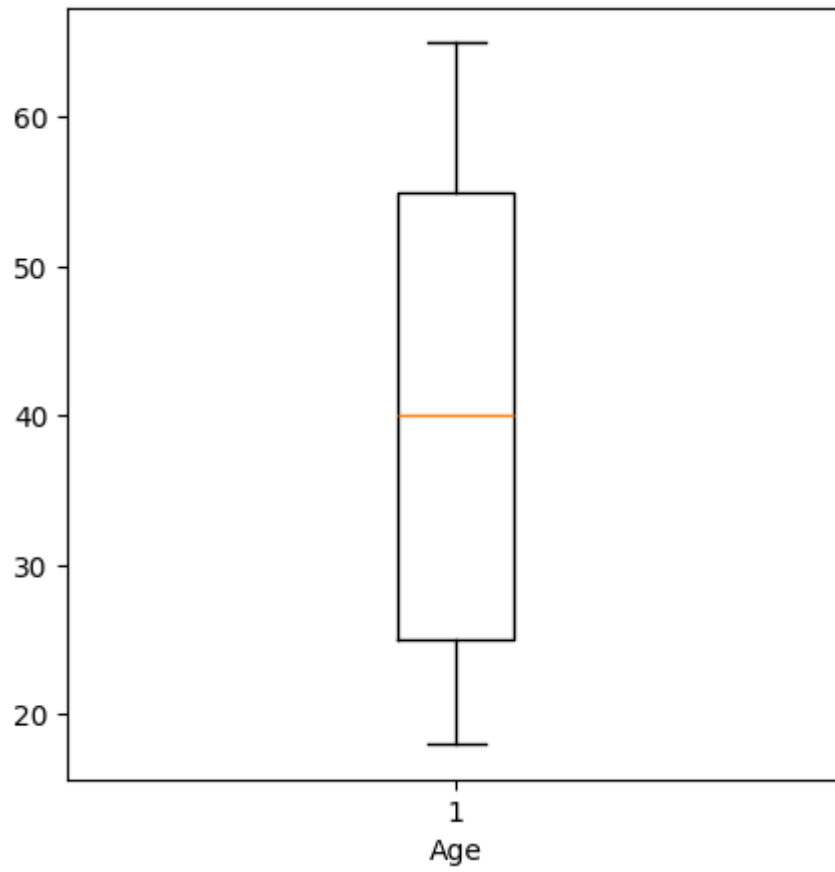
|       | Co-Applicant | Property Price | Loan Sanction Amount (USD) |
|-------|--------------|----------------|----------------------------|
| count | 30000.000000 | 3.000000e+04   | 29660.000000               |
| mean  | -4.743867    | 1.317597e+05   | 47649.342208               |
| std   | 74.614593    | 9.354955e+04   | 48221.146686               |
| min   | -999.000000  | -9.990000e+02  | -999.000000                |
| 25%   | 1.000000     | 6.057216e+04   | 0.000000                   |
| 50%   | 1.000000     | 1.099936e+05   | 35209.395000               |
| 75%   | 1.000000     | 1.788807e+05   | 74261.250000               |
| max   | 1.000000     | 1.077967e+06   | 481907.320000              |

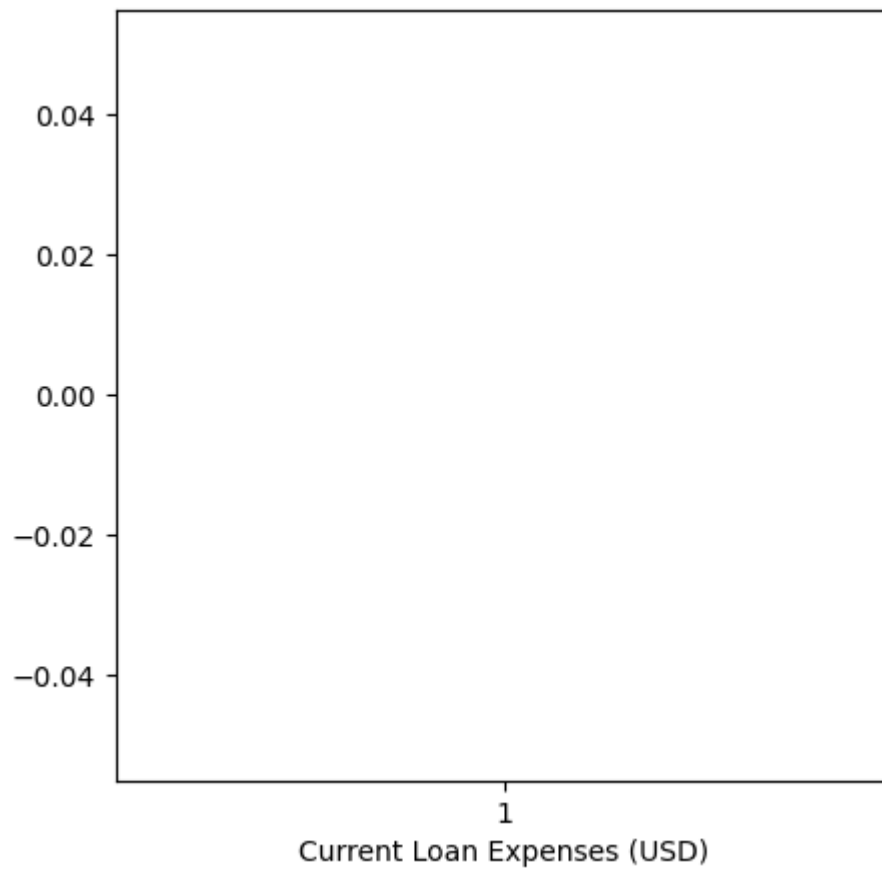
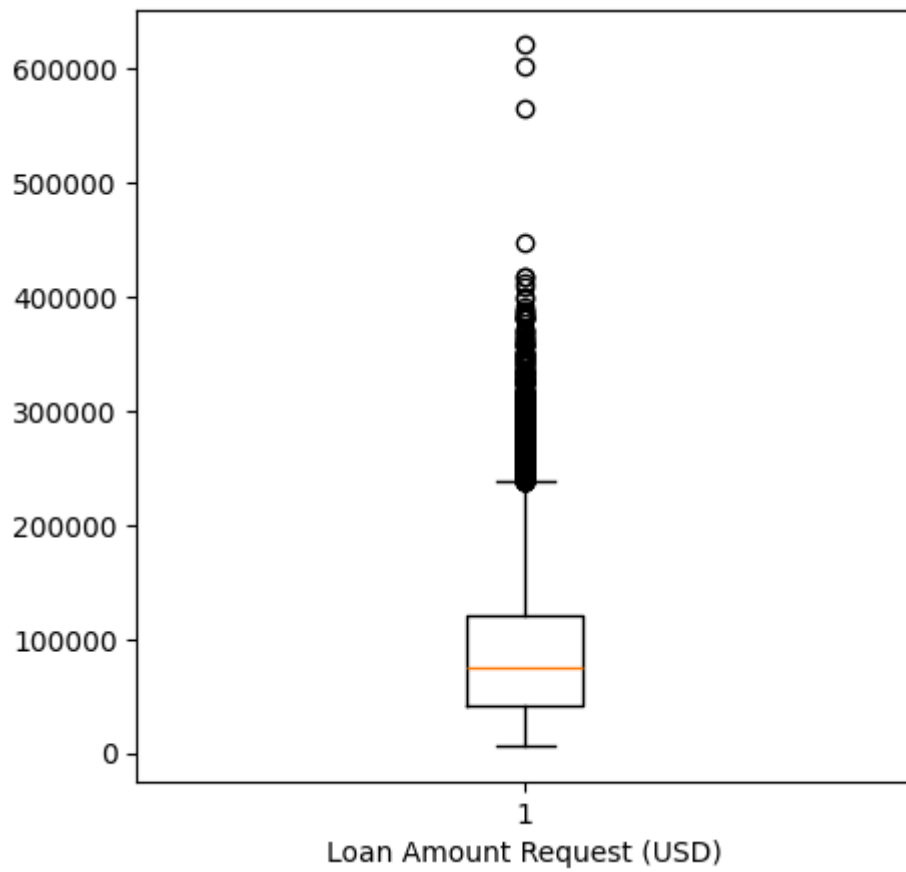
## SKEWNESS

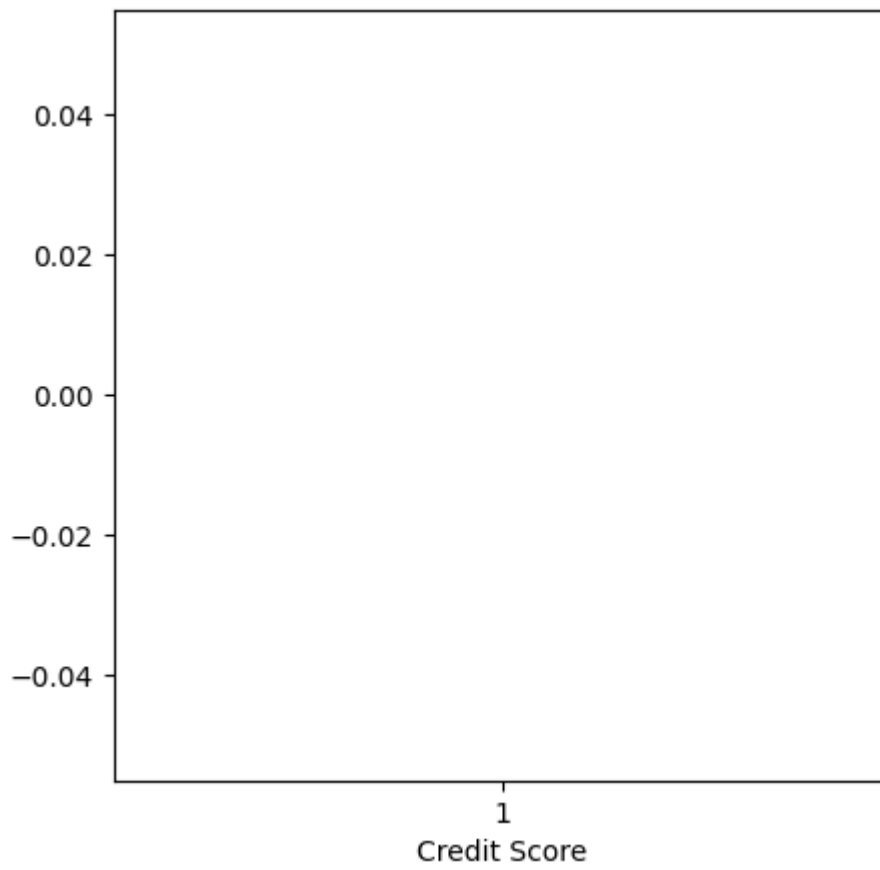
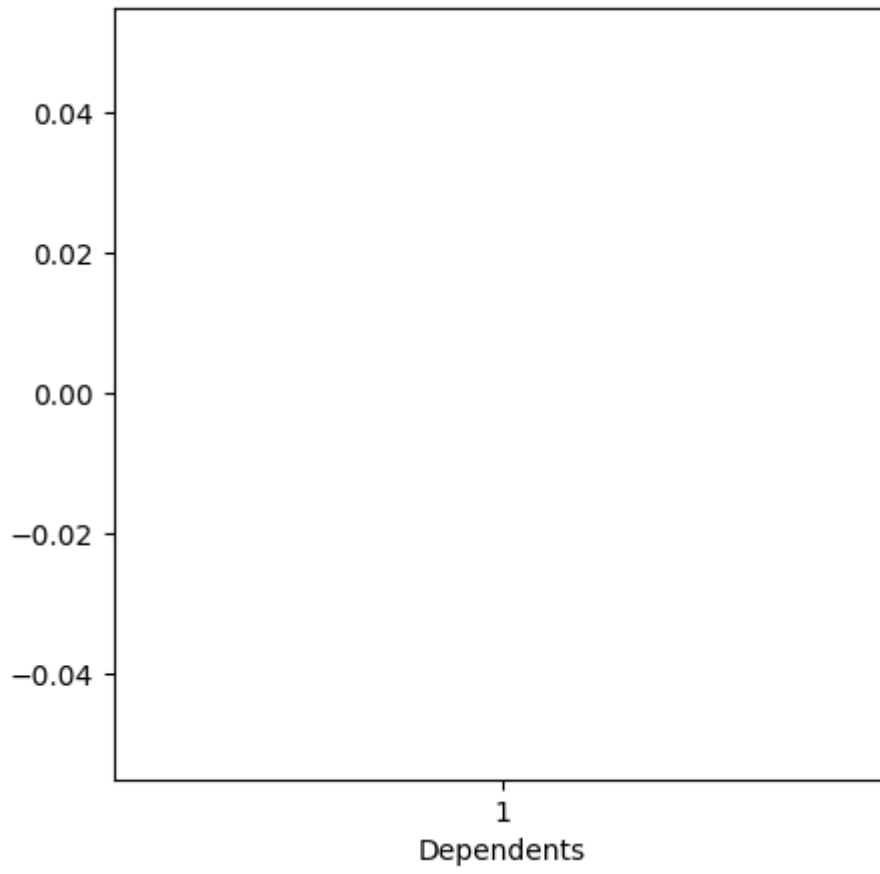
Age:0.046093801598670575  
 Income (USD):154.0017199982858  
 Loan Amount Request (USD):1.2603921833589773  
 Current Loan Expenses (USD):0.04432815526113482  
 Dependents:0.8047511636765929  
 Credit Score:-0.020255169213215216  
 No. of Defaults:1.5483032763275384  
 Property ID:-0.010519189724274059  
 Property Age:153.2196101099395  
 Property Type:0.051070209039890846  
 Co-Applicant:-13.250767889589497  
 Property Price:1.4169646295074756  
 Loan Sanction Amount (USD):1.2299390309619023

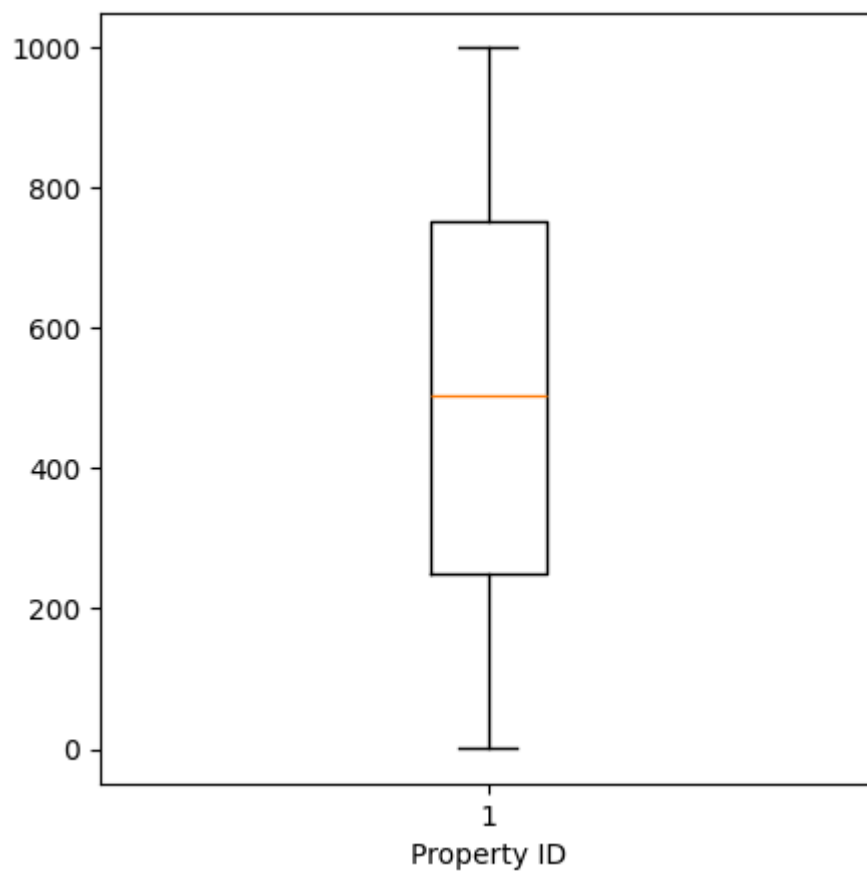
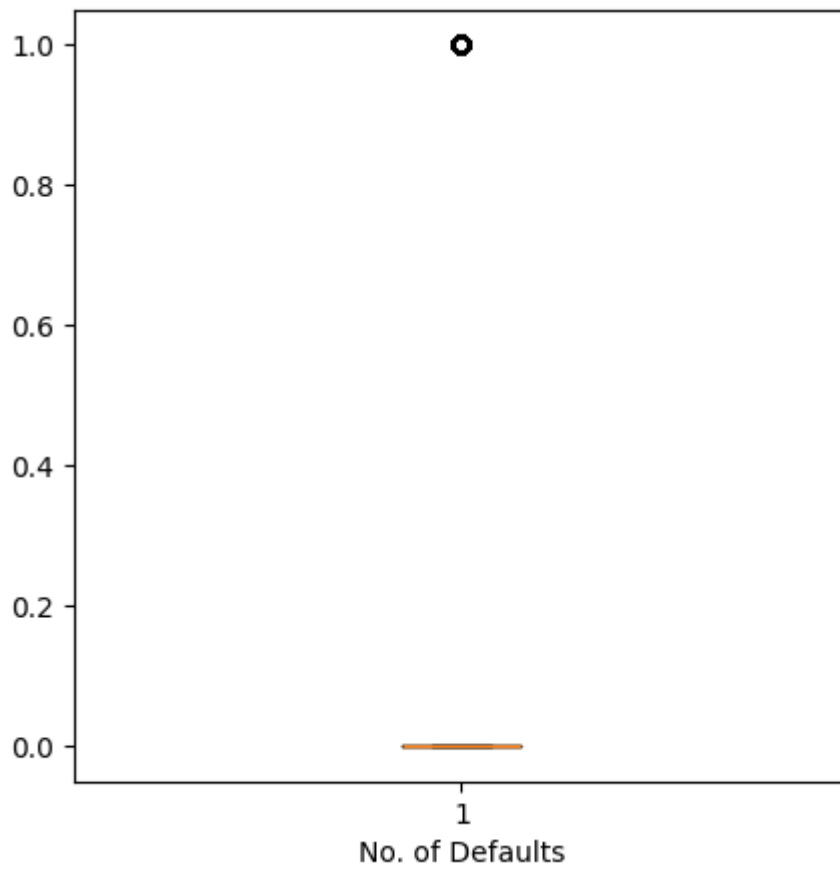


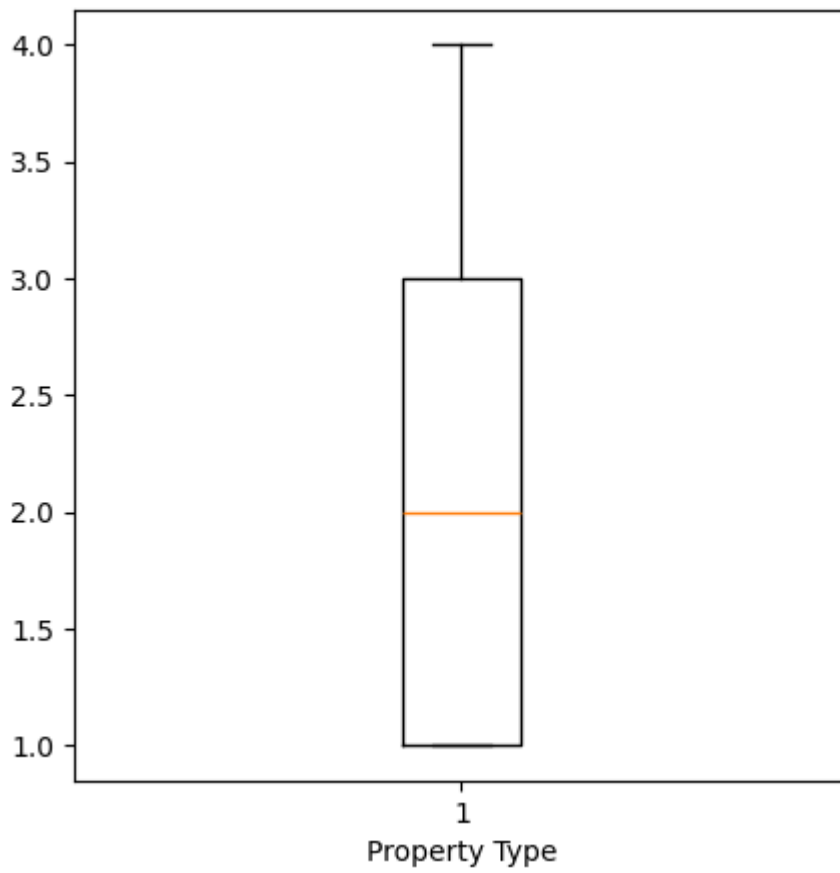
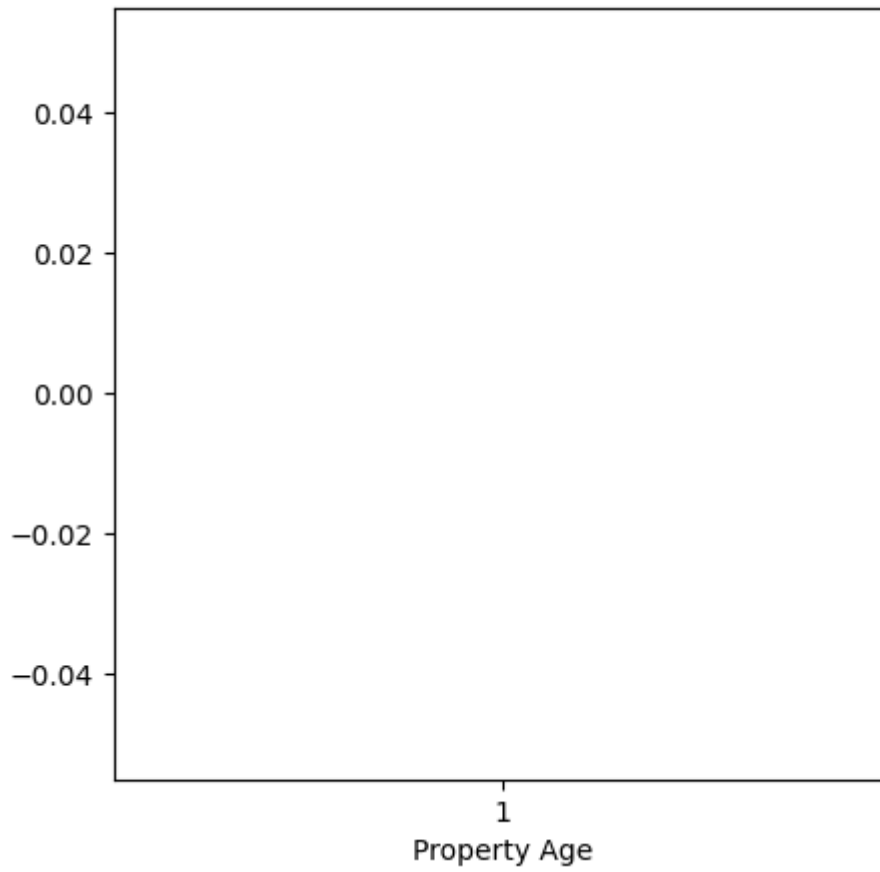
OUTLIERS VISUALS

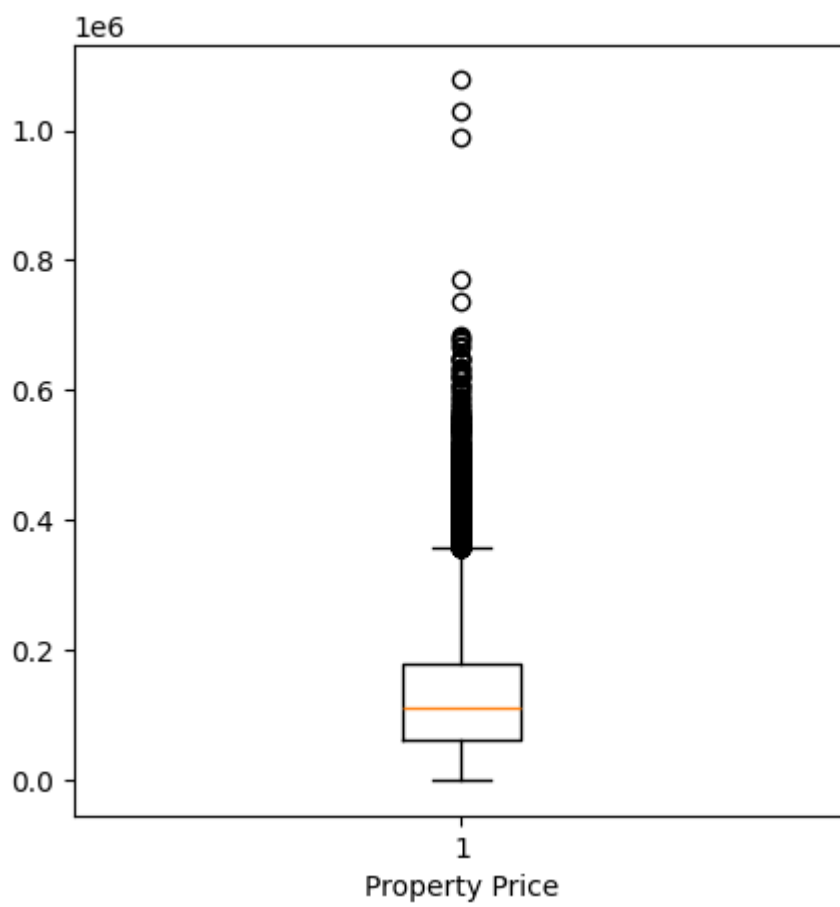
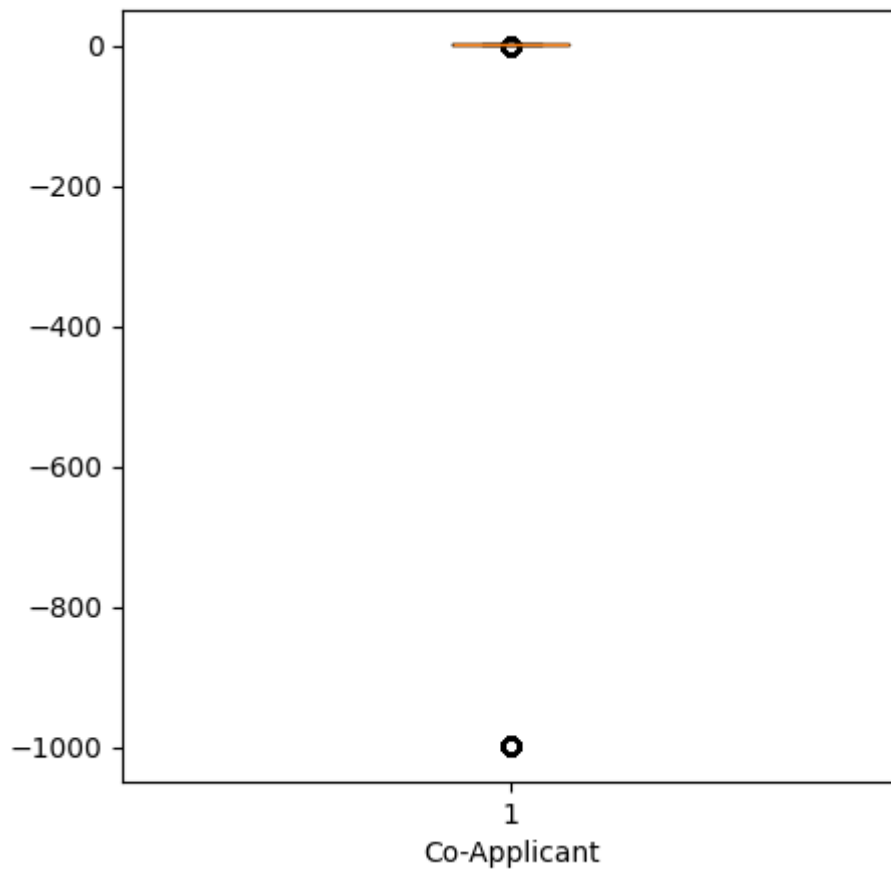


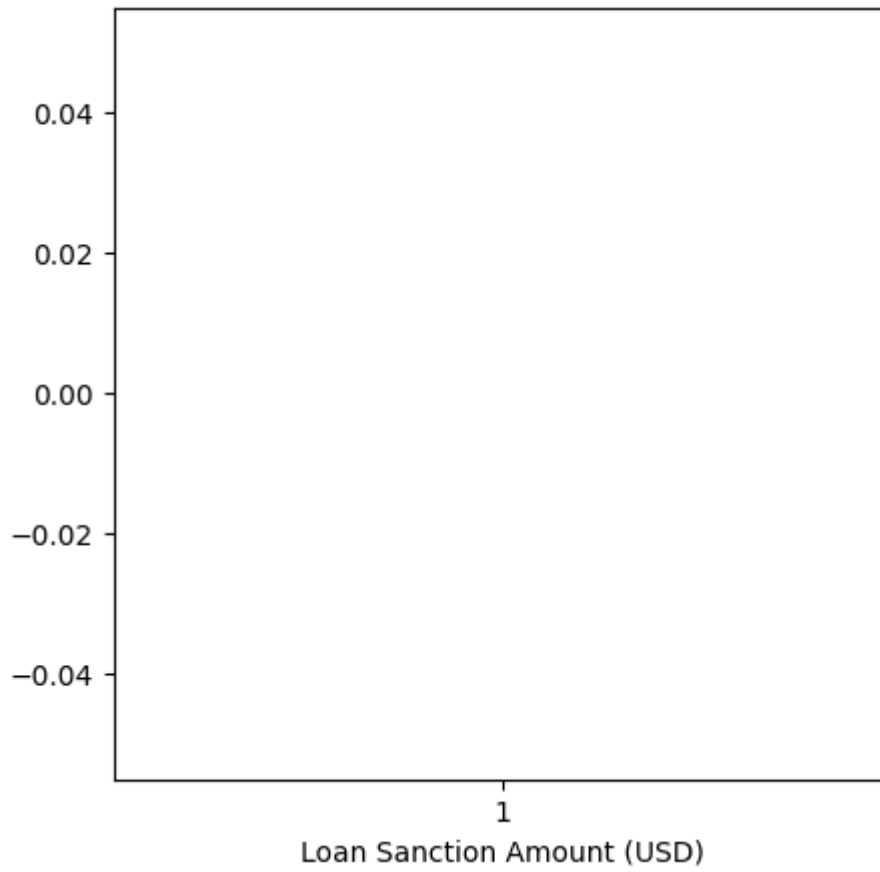












In [ ]:

## Inference Table

| S.No | Observation           | Inference                                                               |
|------|-----------------------|-------------------------------------------------------------------------|
| 1    | Datasets imported     | Enables baseline exploratory data analysis (numerical and categorical). |
| 2    | Histogram plot        | Visualizes feature distributions across the dataset.                    |
| 3    | Box plot              | Identifies outliers in numerical features.                              |
| 4    | Correlation heatmap   | Evaluates linear relationships and feature collinearity.                |
| 5    | Missing value heatmap | Visualizes missing data patterns across columns.                        |

## Refelction on Learning Outcomes

- Learn to use python libraries.
- Learn how to explore dataset and use required plots to understand the data